Operations Research 1: Deterministic Models
Deterministic Dynamic Programming

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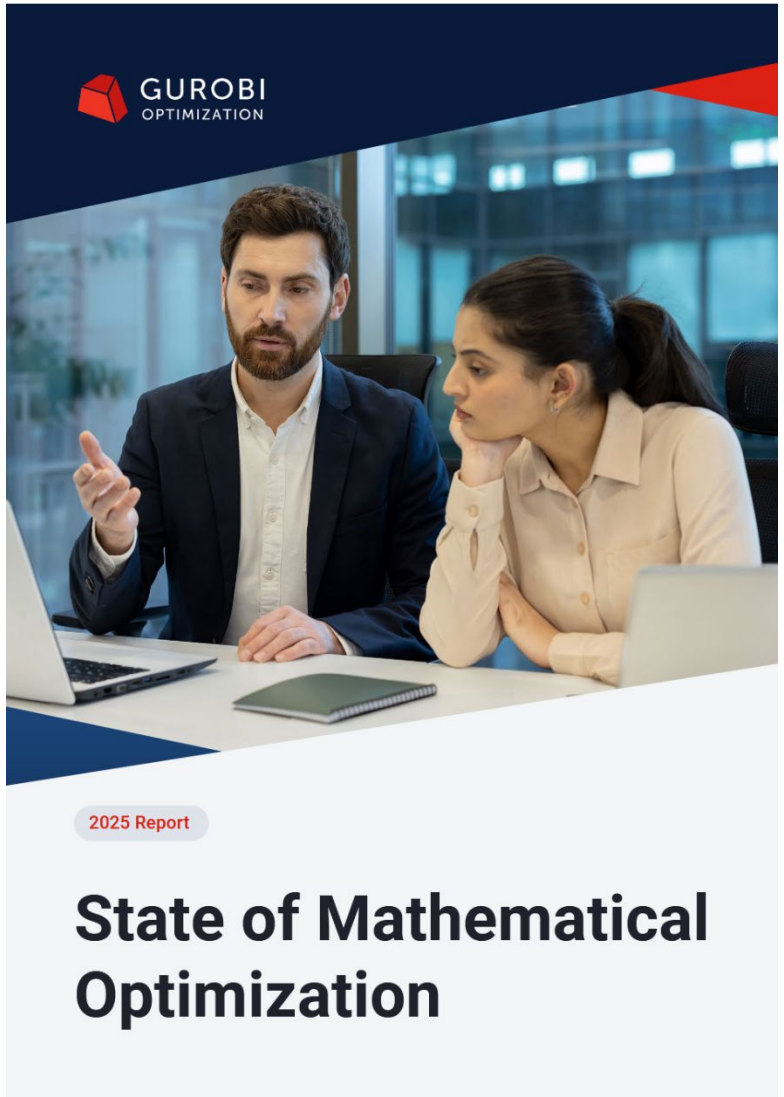
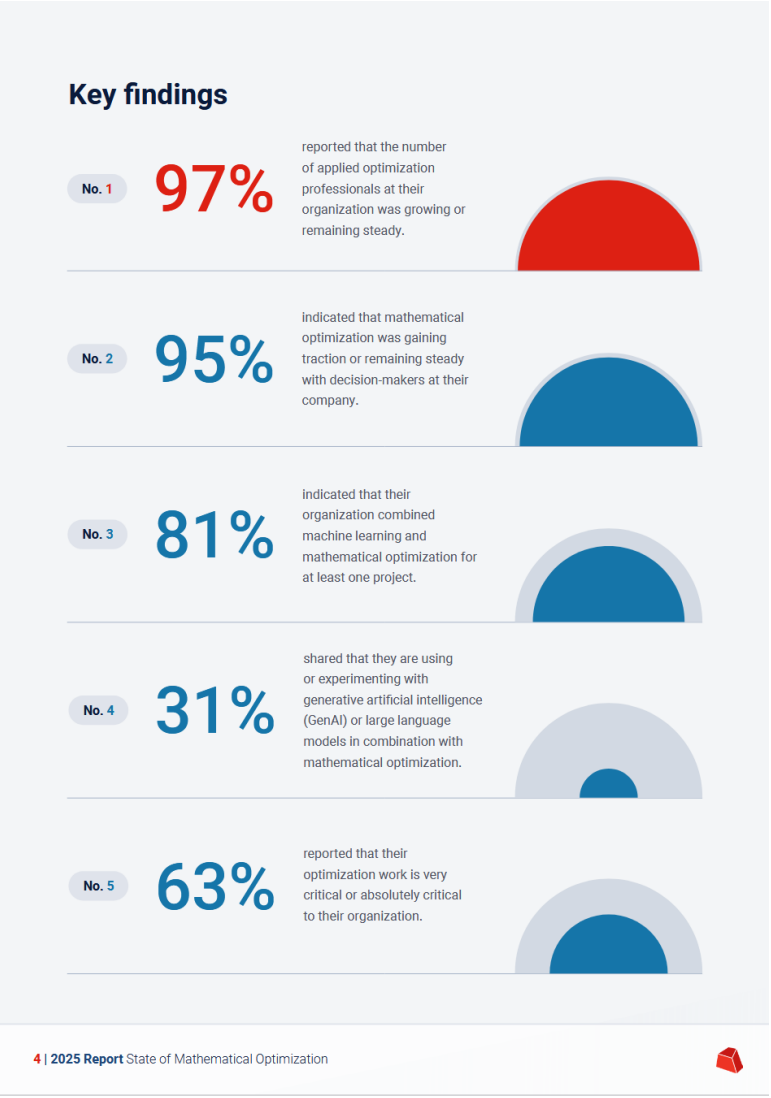
Operations & Services Research (TOpS) Laboratory
Department of Industrial Engineering
Tsinghua University, China

Homework 02: LP Modeling (with AI Tools)

Gurobi: State of Mathematical Optimization (2025)





Key findings

No. 1 **97%** reported that the number of applied optimization professionals at their organization was growing or remaining steady.

No. 2 **95%** indicated that mathematical optimization was gaining traction or remaining steady with decision-makers at their company.

No. 3 **81%** indicated that their organization combined machine learning and mathematical optimization for at least one project.

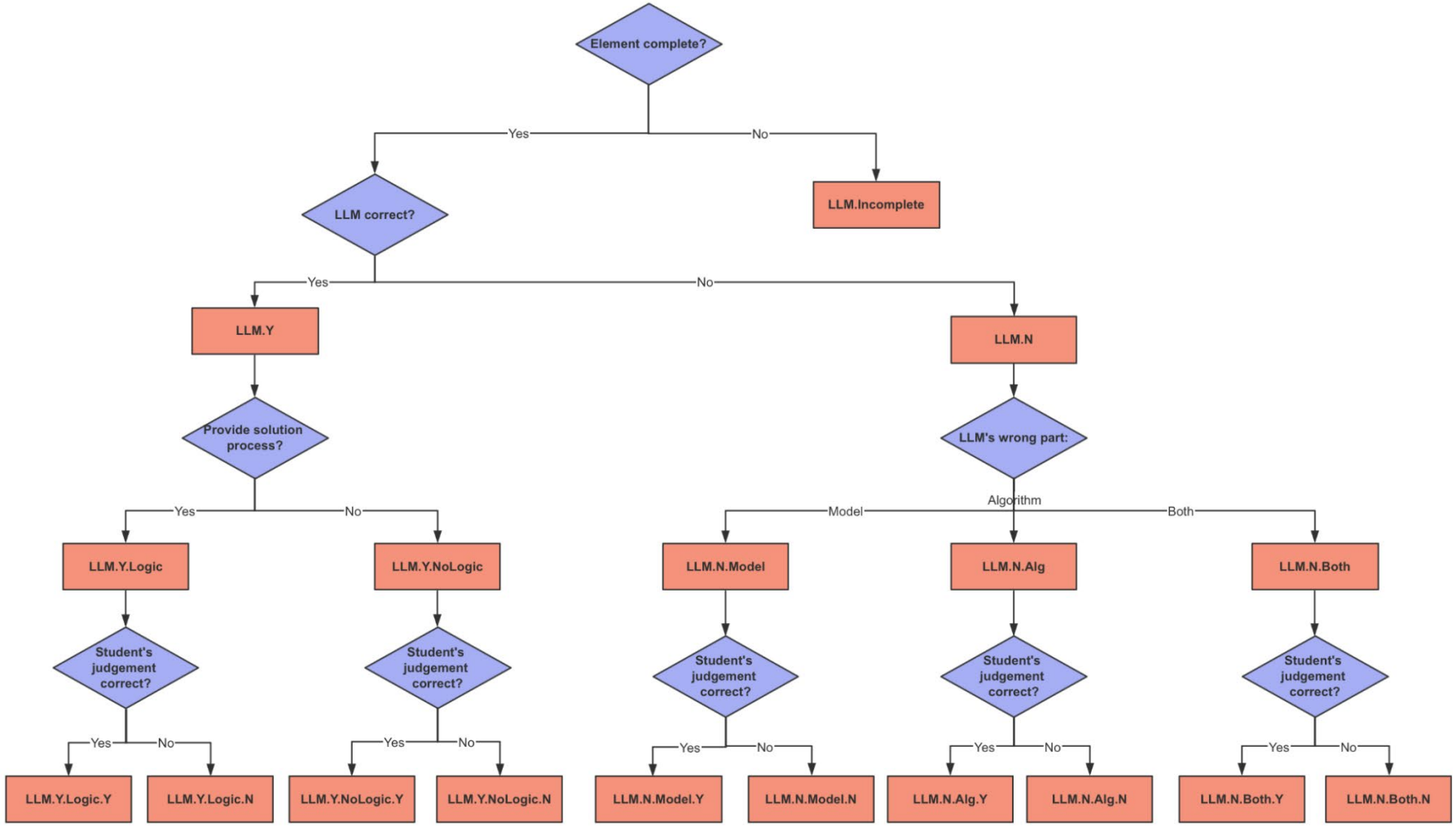
No. 4 **31%** shared that they are using or experimenting with generative artificial intelligence (GenAI) or large language models in combination with mathematical optimization.

No. 5 **63%** reported that their optimization work is very critical or absolutely critical to their organization.

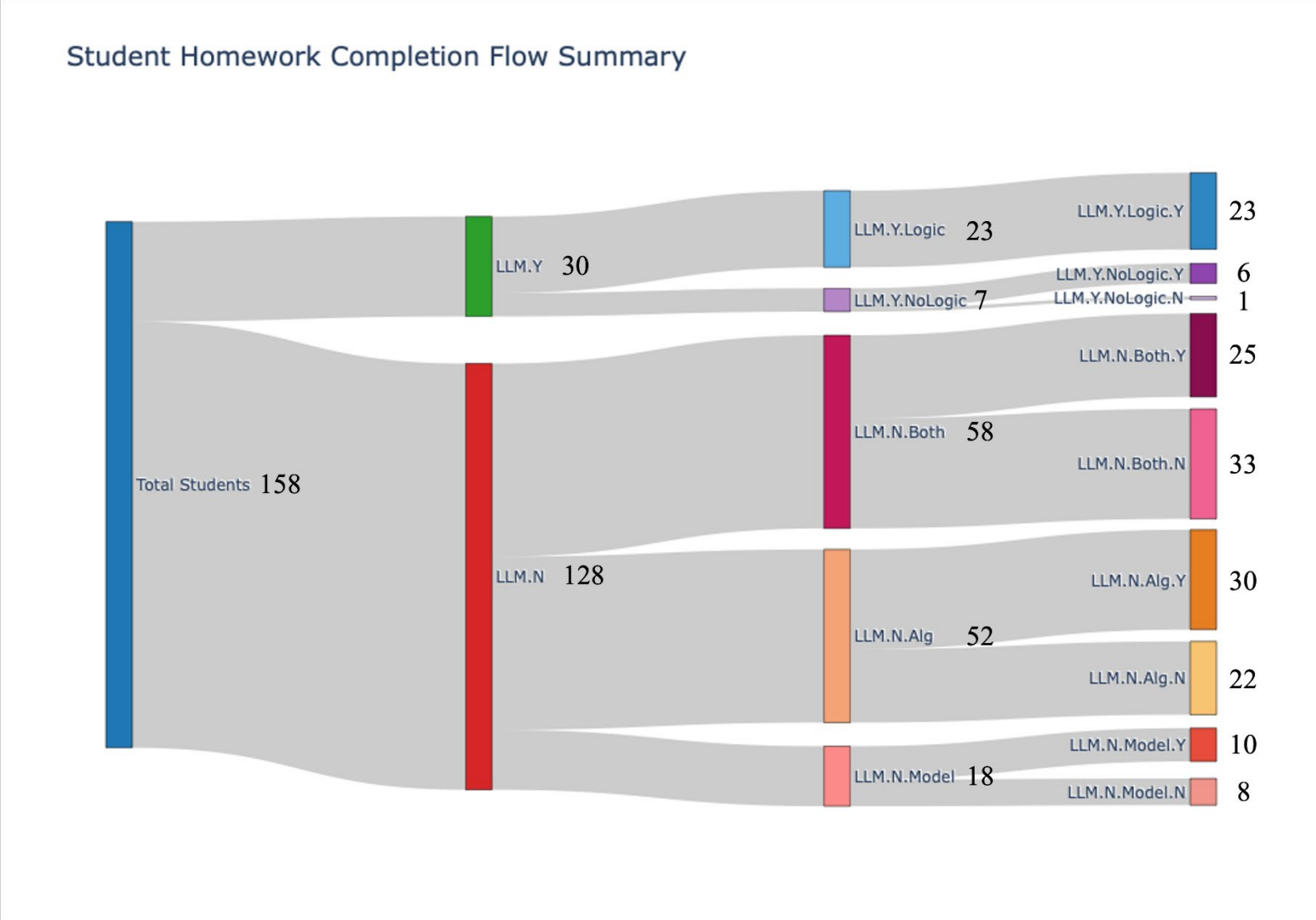
4 | 2025 Report State of Mathematical Optimization

Source Gurobi (2025)

Homework 02: Flow Chart



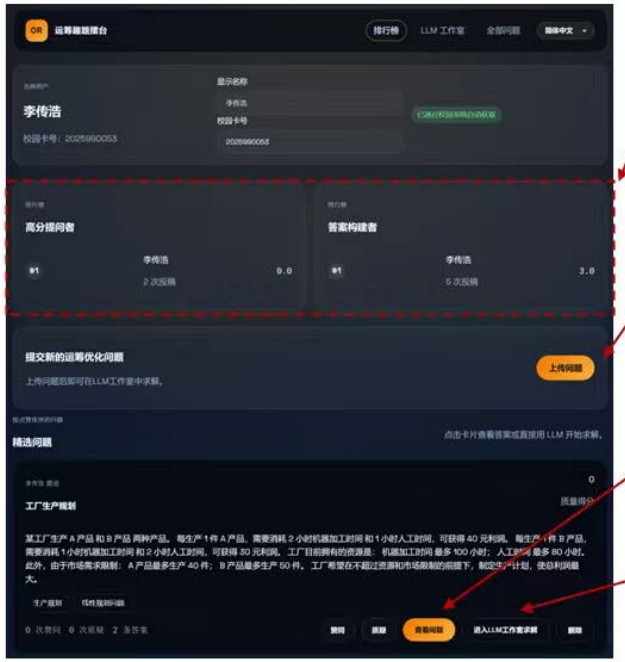
Homework 02: Sankey Diagram



运筹趣题擂台

1. 登录
账号：校园卡号
默认密码：iisebok$888

2. 点击进入运筹趣题擂台
遇到“您要提交的信息不安全”，点“仍然发送”



3. 运筹趣题擂台页面说明

3.1 排行榜
提问榜（回答榜）按照用户上传的所有问题（回答）获得的总upvotes-downvotes数来排列，最多显示前10名

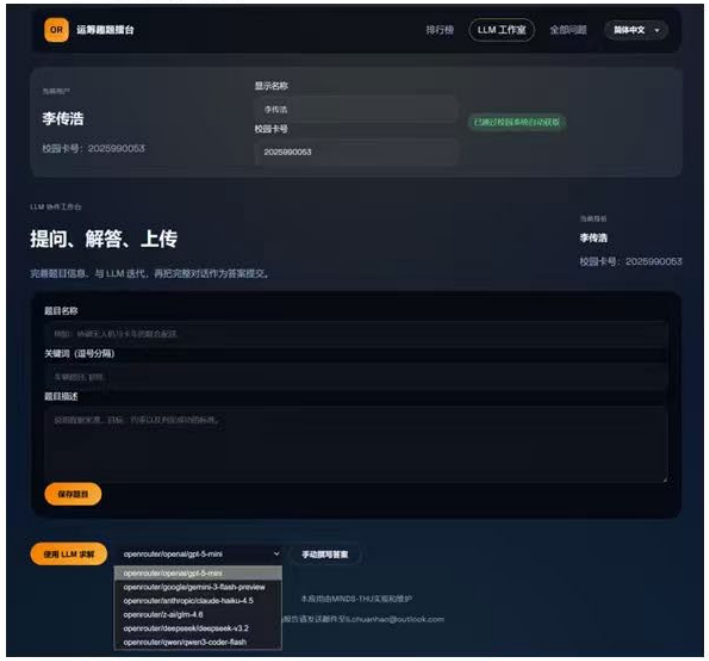
3.2 提交新问题
点击“上传问题”会进入“LLM工作室页面”，在表格中填写问题名称、关键词和问题描述，点击保存题目即可完成上传

3.3 查看现有问题和对应回答
在问题列表中点击“查看问题”会进入该问题页面，可以查看详细问题描述，以及用户名上传的回答
点击进入“进入LLM工作室求解”，会进入“LLM工作室页面”

运筹趣题擂台使用说明

学科知识引擎入口：
https://www2.ie.tsinghua.edu.cn/iisebok/index.php

LLM工作室页面



两种解答问题方式：
1. 与LLM对话来解题，然后把整个对话作为答案上传
2. 手动撰写回答，然后上传

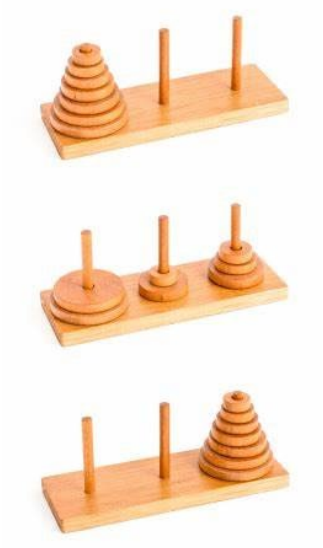
End
12月30日
（周二）
23:59

Link: https://www2.ie.tsinghua.edu.cn/iisebok/login.php

Introduction

Deterministic Dynamic Programming

Rubik's cube



Tower of Hanoi

Chinese nine linked ring puzzle

Huarong Pass sliding block puzzle

Source: https://www.rubiks.com, https://mathworld.wolfram.com/TowerofHanoi.html, https://chinesepuzzles.org, Internet

Deterministic Finite State Finite Horizon Problem

Deterministic Finite State Finite Horizon Problem

- In all dynamic programming (DP) problems, the central object is a discrete-time dynamic system that generates a sequence of states under the influence of actions or controls.

- The system may evolve deterministically (this course) or randomly (under additional influence of a random disturbance)

Source: Bertsekas (2023)

Deterministic Finite State Finite Horizon Problem

- In finite horizon problems
  - A finite number of T time steps, indexed by time t (decision epochs or stages)
  - State of the system S_t at time t
  - Control or decision variable $x_t \in X_t(S_t)$ at time t, which depends on S_t
  - In a deterministic DP problem, the transition function (or system equation) is

$$S_{t+1} = f_t(S_t, x_t),\ t = 0, 1, \dots, T-1.$$

    - That is, the state S_{t+1} is determined solely by state S_t and decision x_t.
  - The system involves an additive cost function $c_t(S_t, x_t)$.
  - Given initial state S_0, the total cost of a control sequence $\{x_0, \dots, x_{T-1}\}$ is

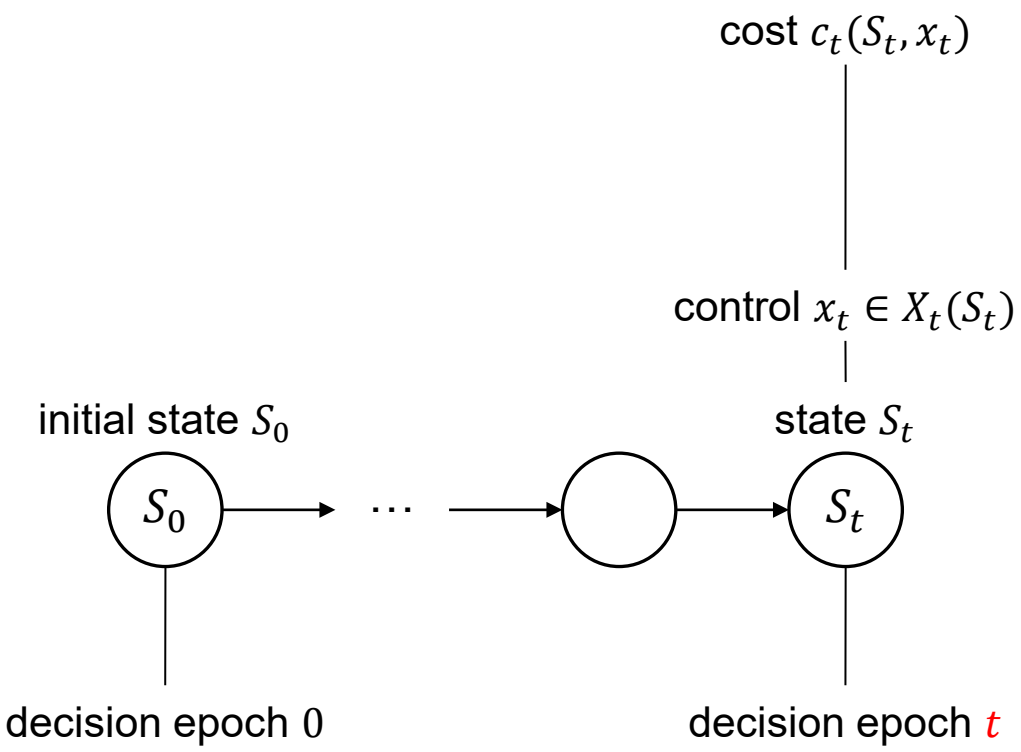
$$V(S_0; x_0, \dots, x_{T-1}) = c_T(S_T) + \sum_{t=0}^{T-1} c_t(S_t, x_t),$$

    - where $c_T(S_T)$ is the terminal cost of terminal state S_T.
    - $V(S_0; x_0, \dots, x_{T-1})$ is well-defined, since $\{x_0, \dots, x_{T-1}\}$ and S_0 determines exactly $\{S_1, \dots, S_T\}$.
  - Minimize the total cost over all feasible sequences: optimal value as a function of S_0

$$V^*(S_0) = \min_{x_t \in X_t(S_t),\ t=0,\dots,T-1} V(S_0; x_0, \dots, x_{T-1}).$$

Source: Modified from Bertsekas (2023) and Powell (2022)

*Deterministic Finite State Finite Horizon Problem



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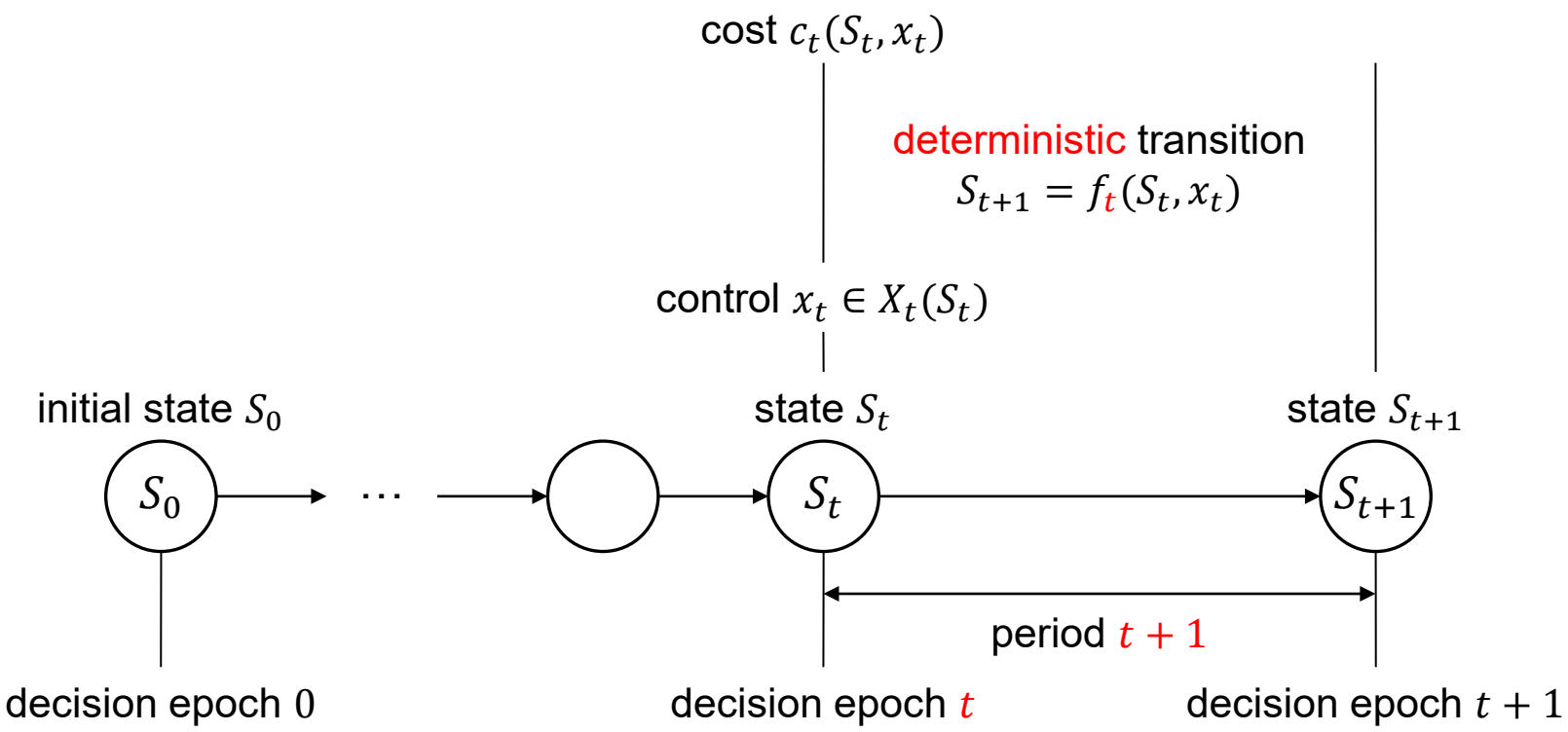

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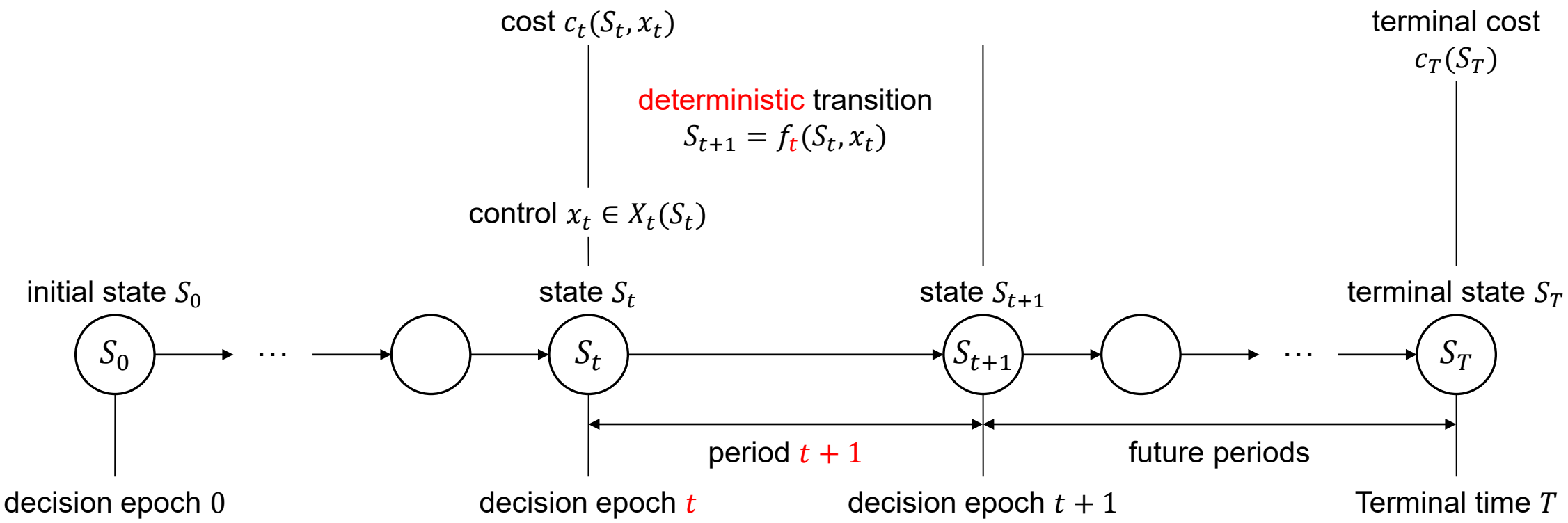

Source: Modified from Bertsekas (2023)

Equivalence to Shortest Path Problem

Deterministic Finite State Finite Horizon Problem

- A deterministic finite state finite horizon problem is equivalent to finding a shortest path from the initial nodes of the (acyclic) graph to the terminal node.

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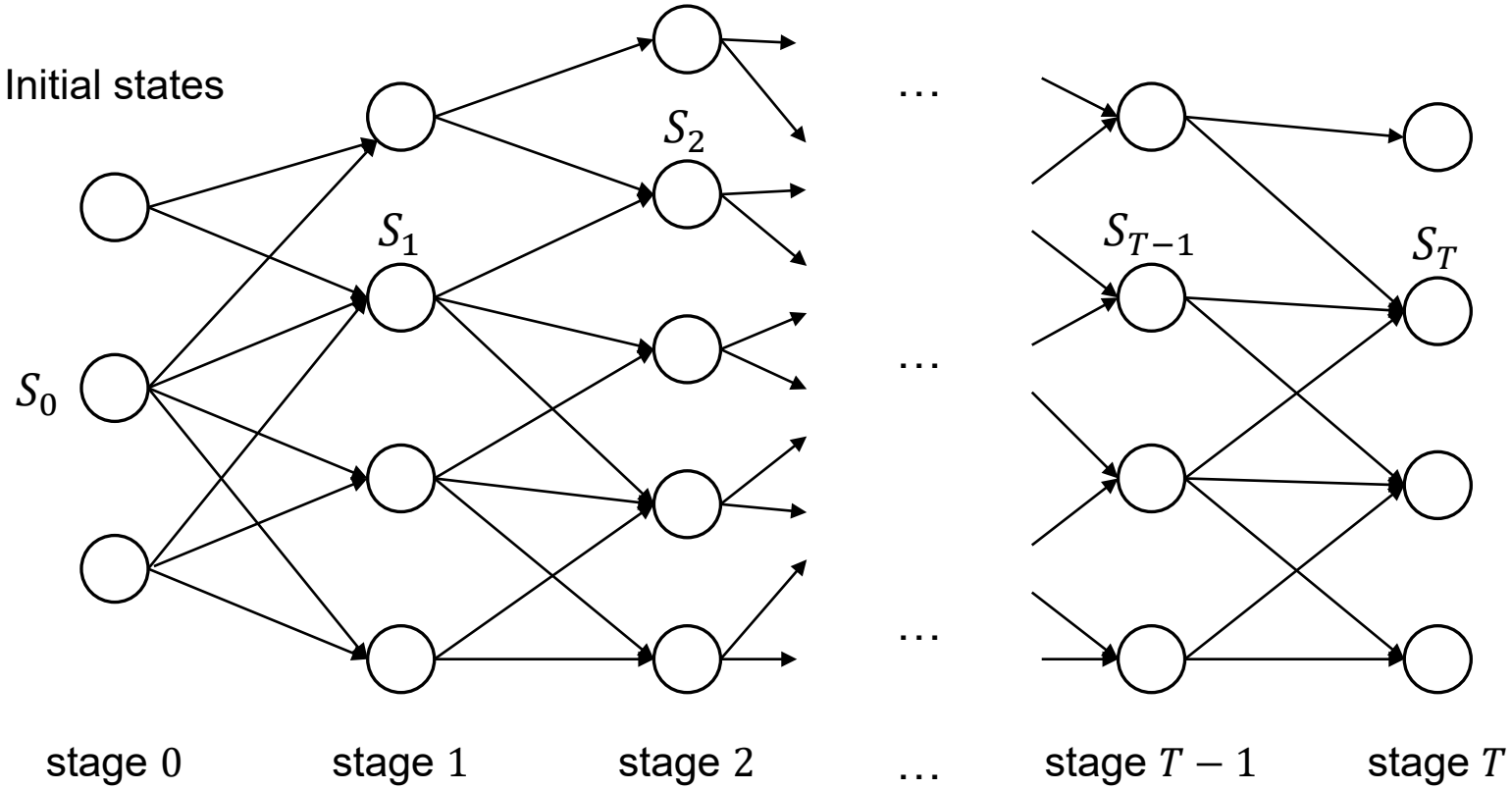

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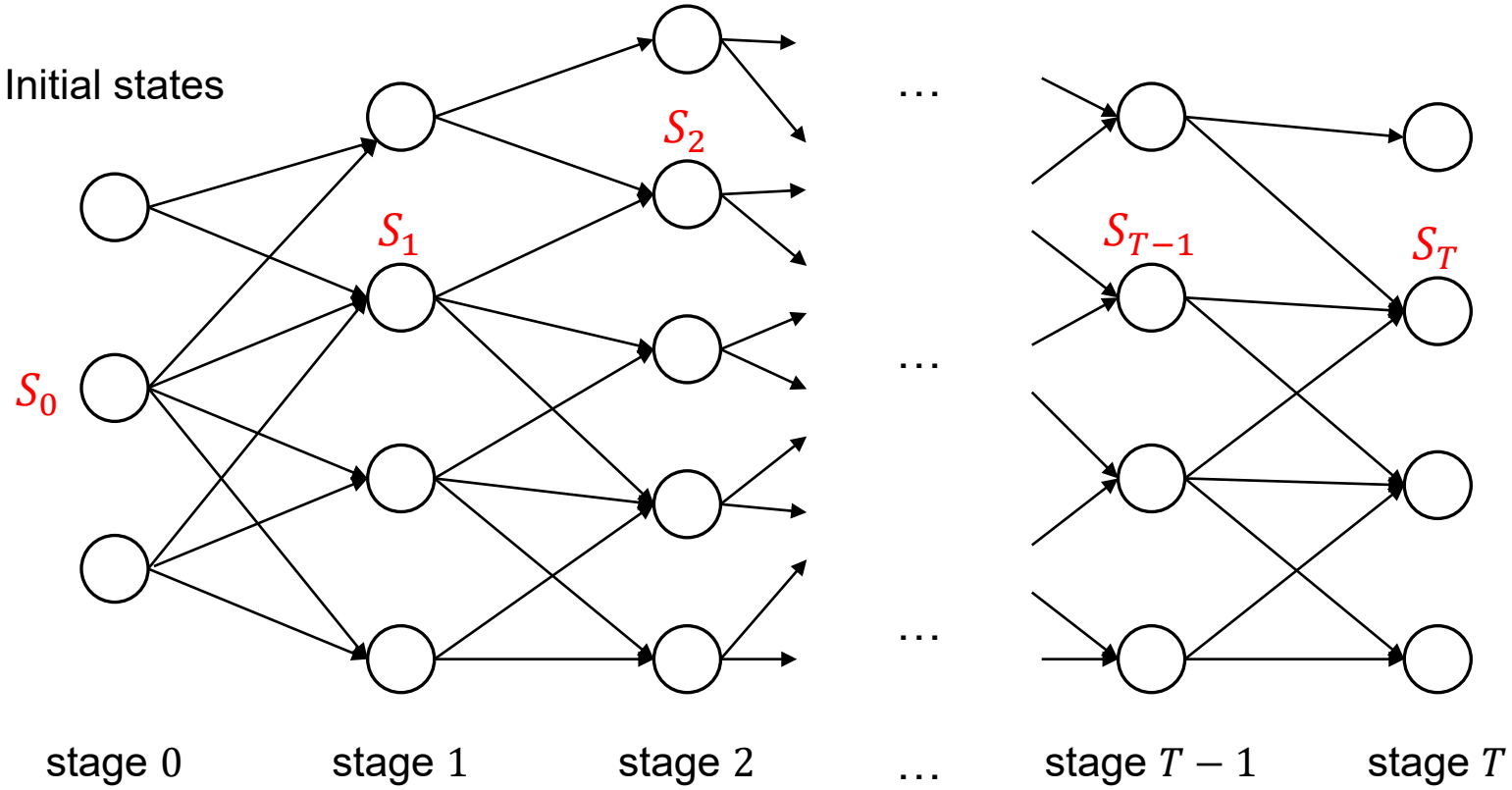

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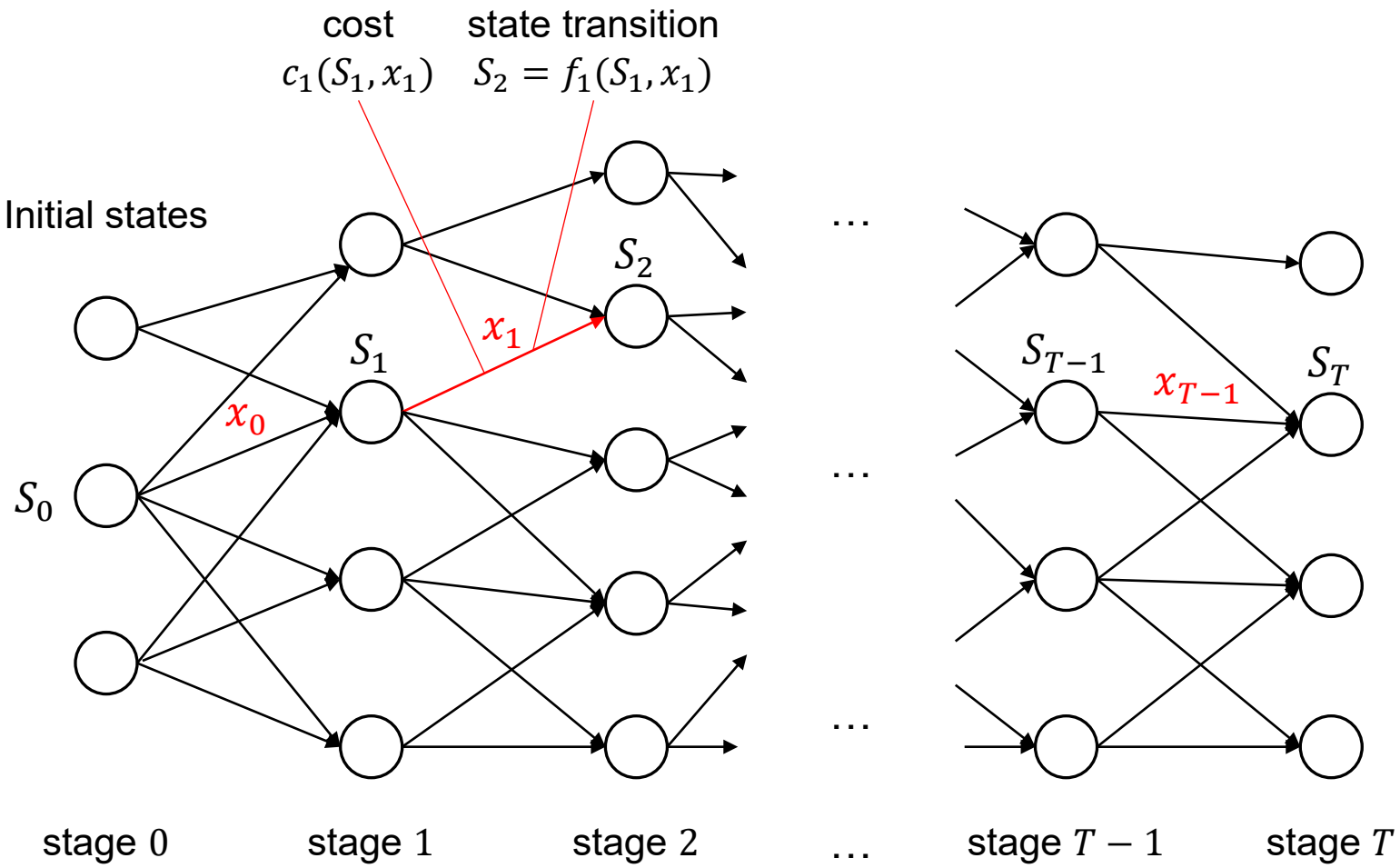

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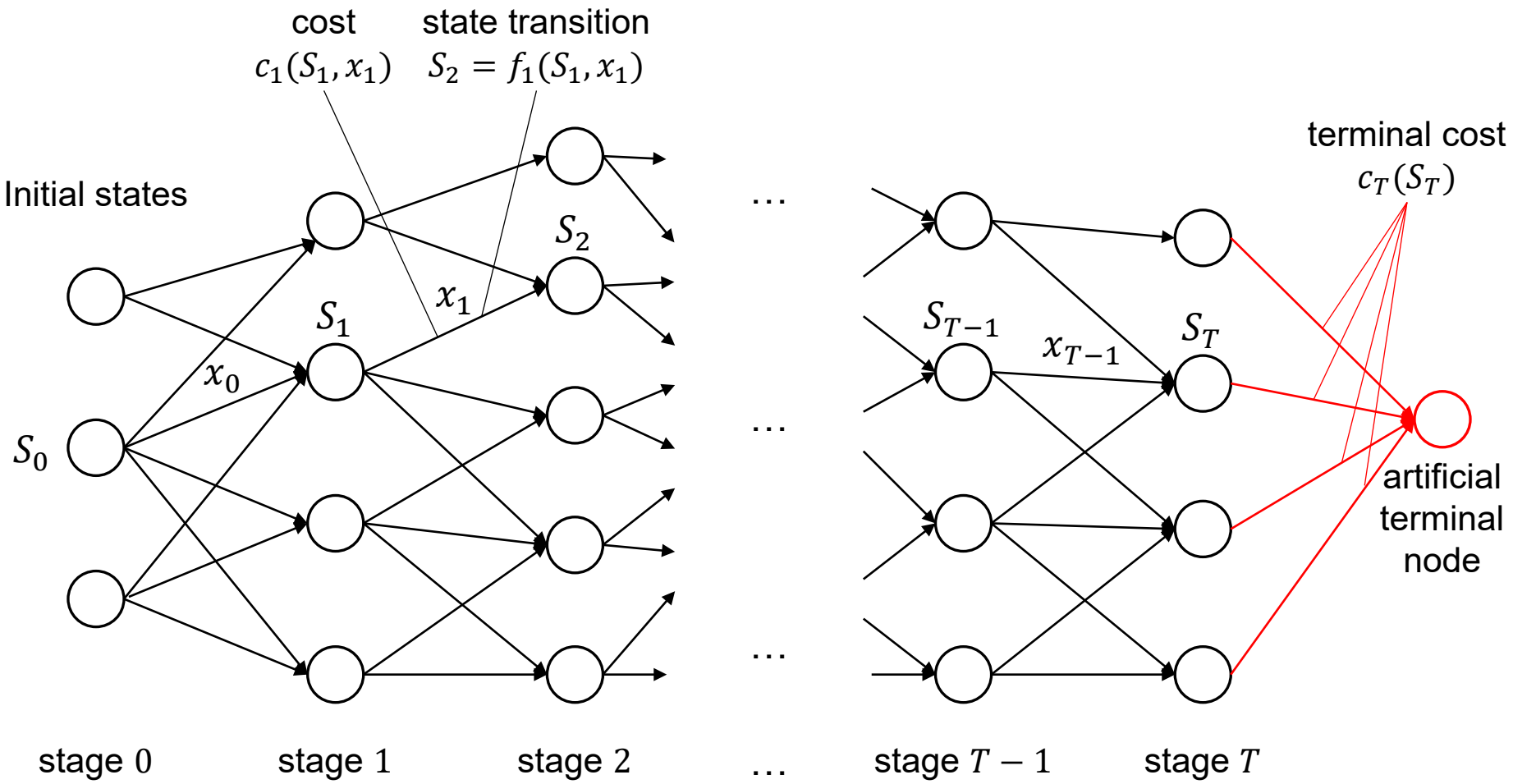

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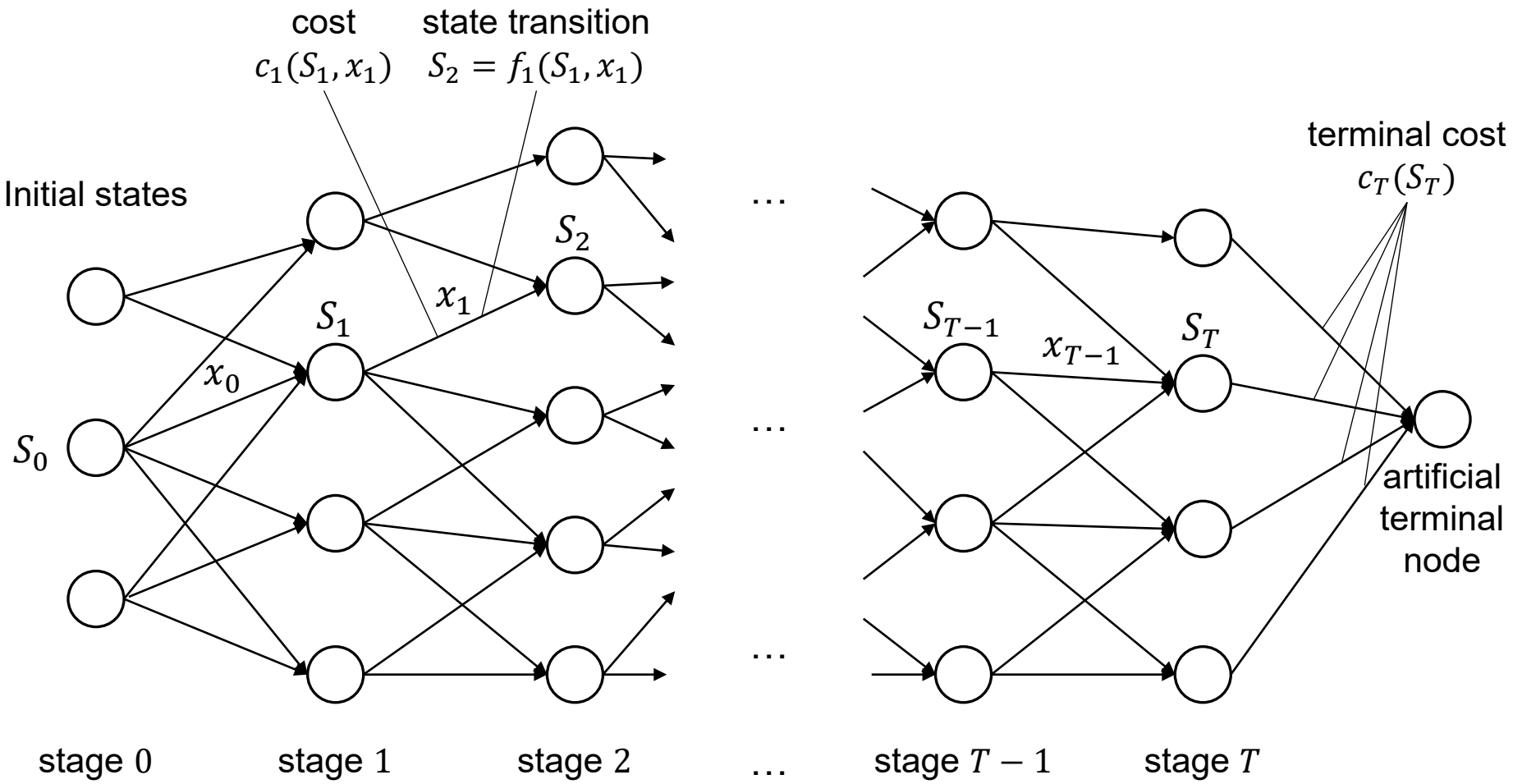

Source: Modified from Bertsekas (2023)

Example: Four Queens Problem

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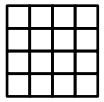
- Place four queens on a 4 x 4 portion of a chessboard so that no queen can attack another.



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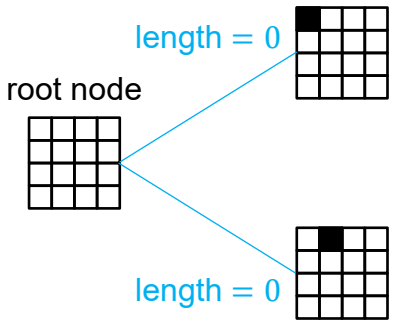

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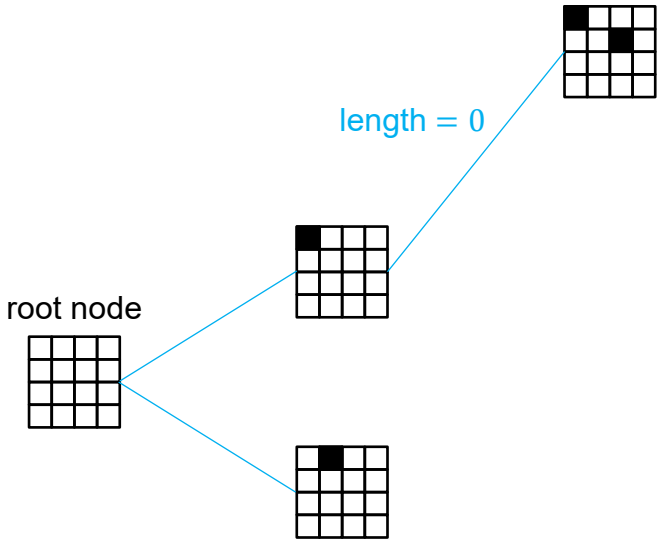

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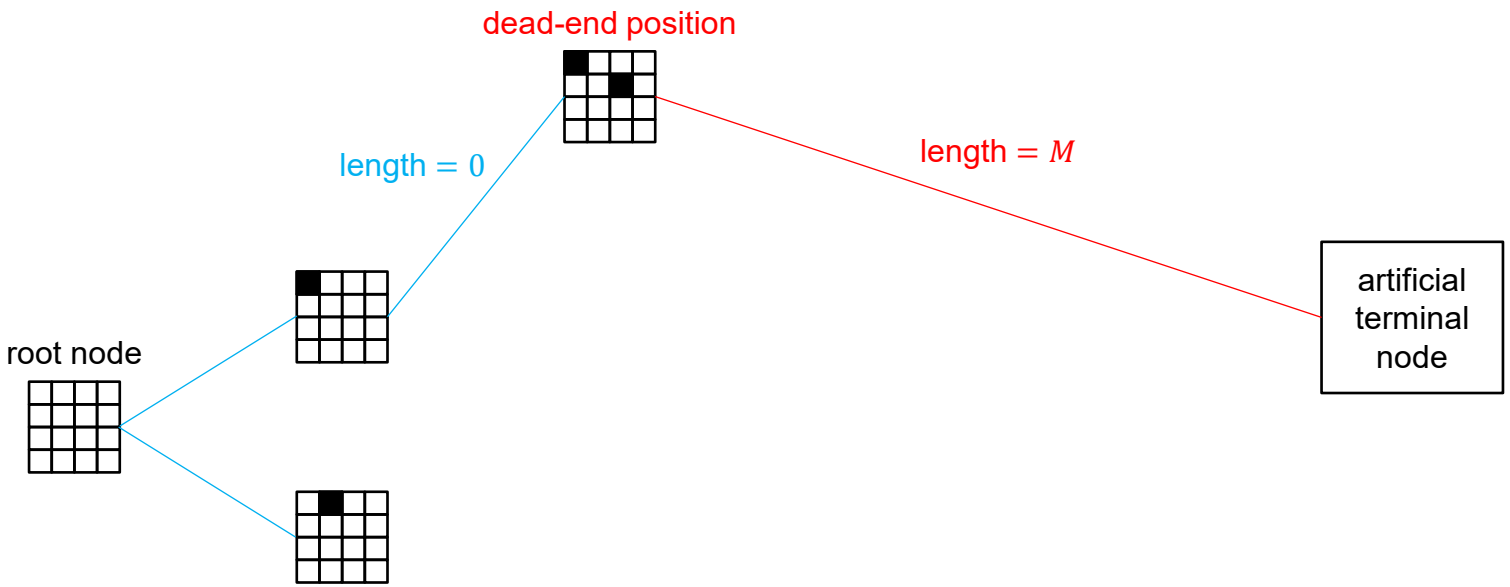

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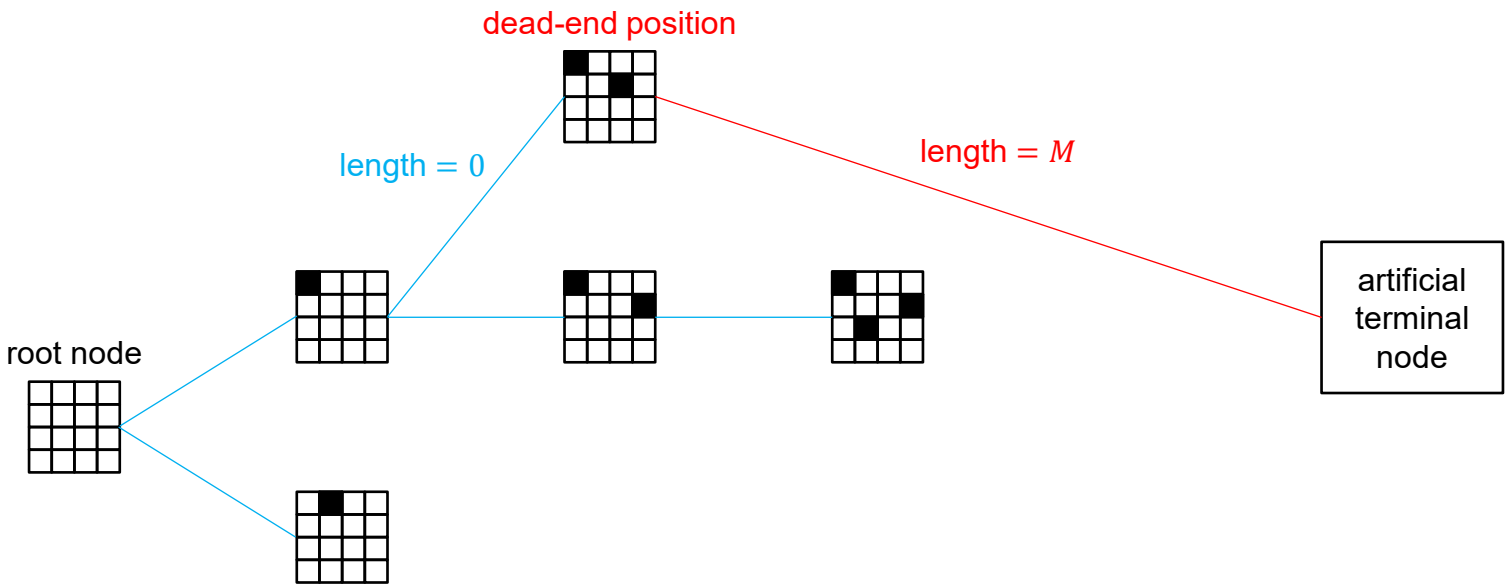

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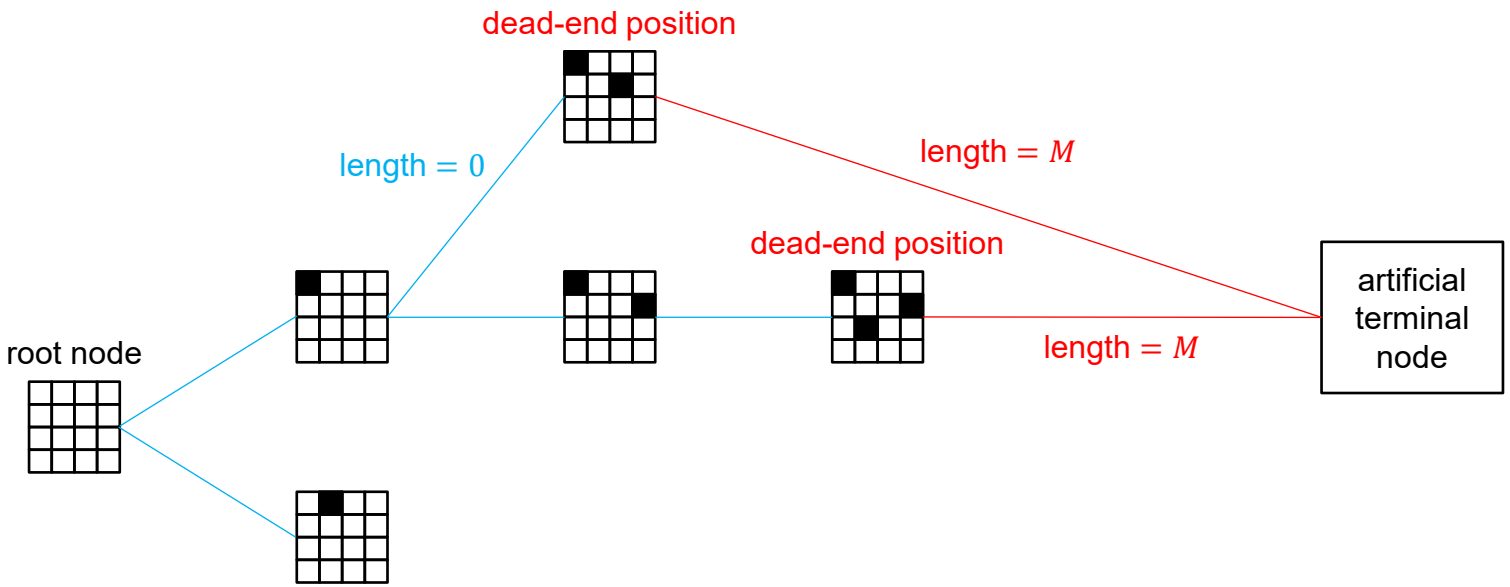

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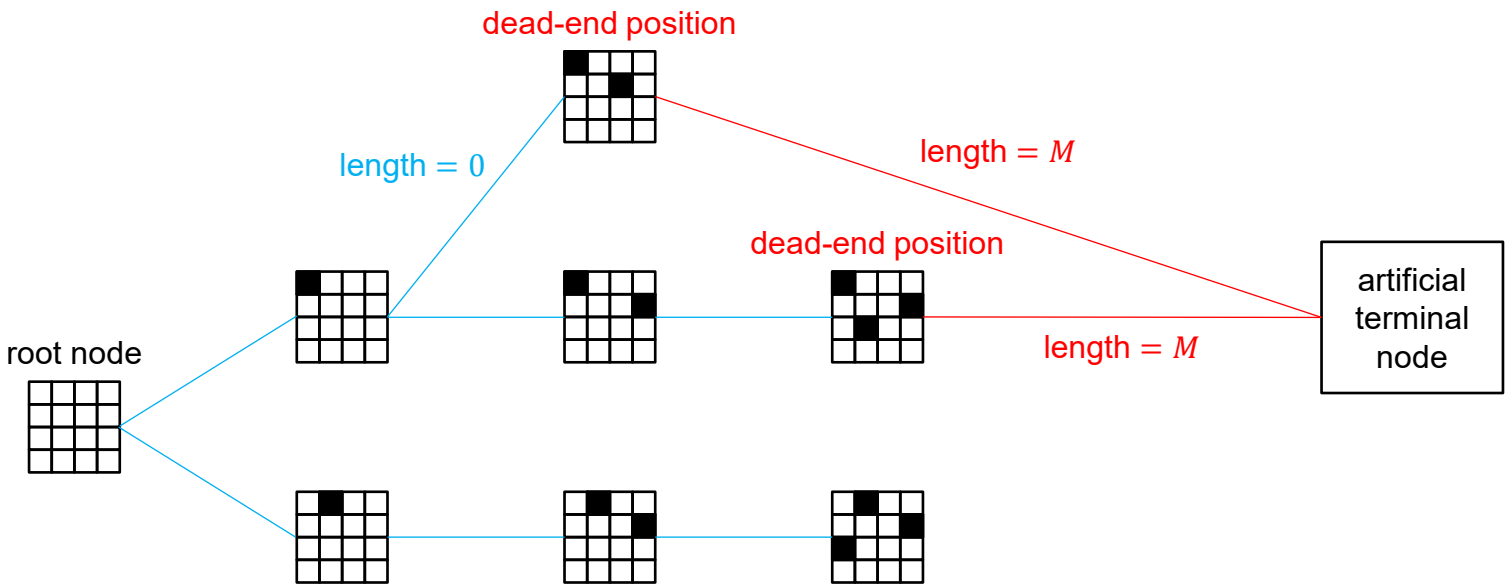

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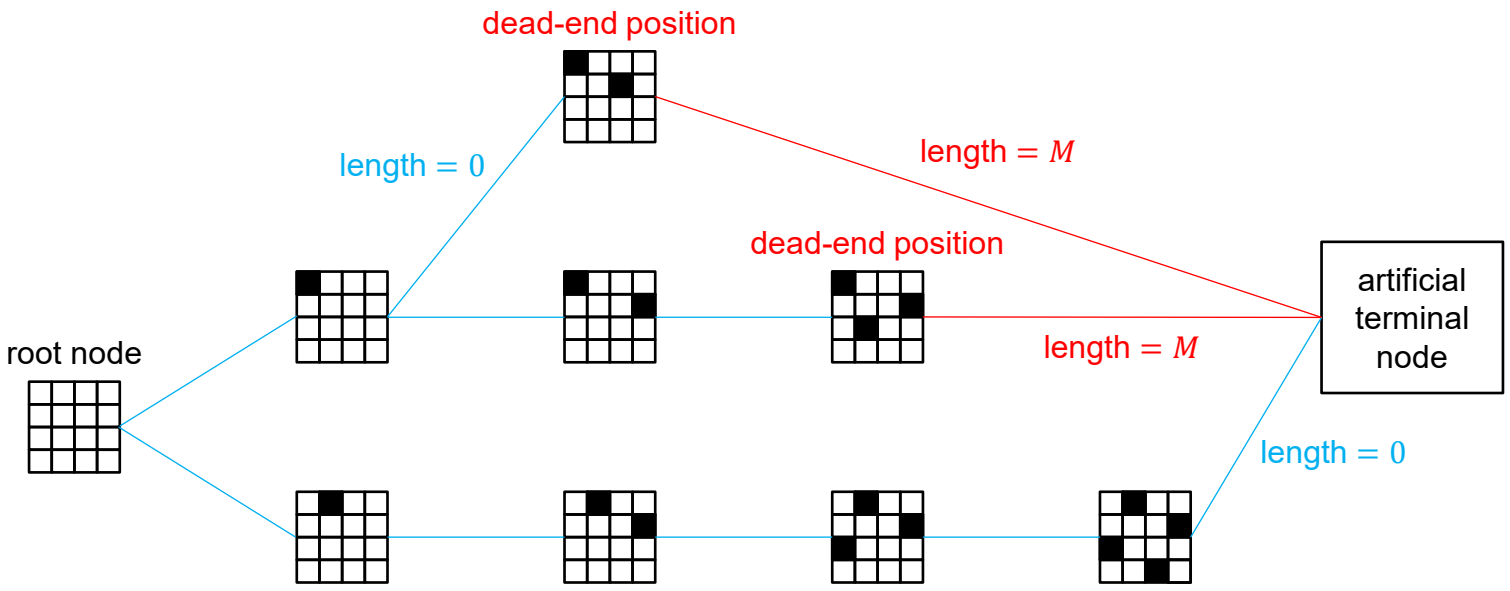


Source: Modified from Bertsekas (2005)

Combinatorial Optimization Problem

Deterministic Finite State Finite Horizon Problem

- Generally, combinatorial optimization problems can be formulated as deterministic finite state finite horizon optimal control problem.

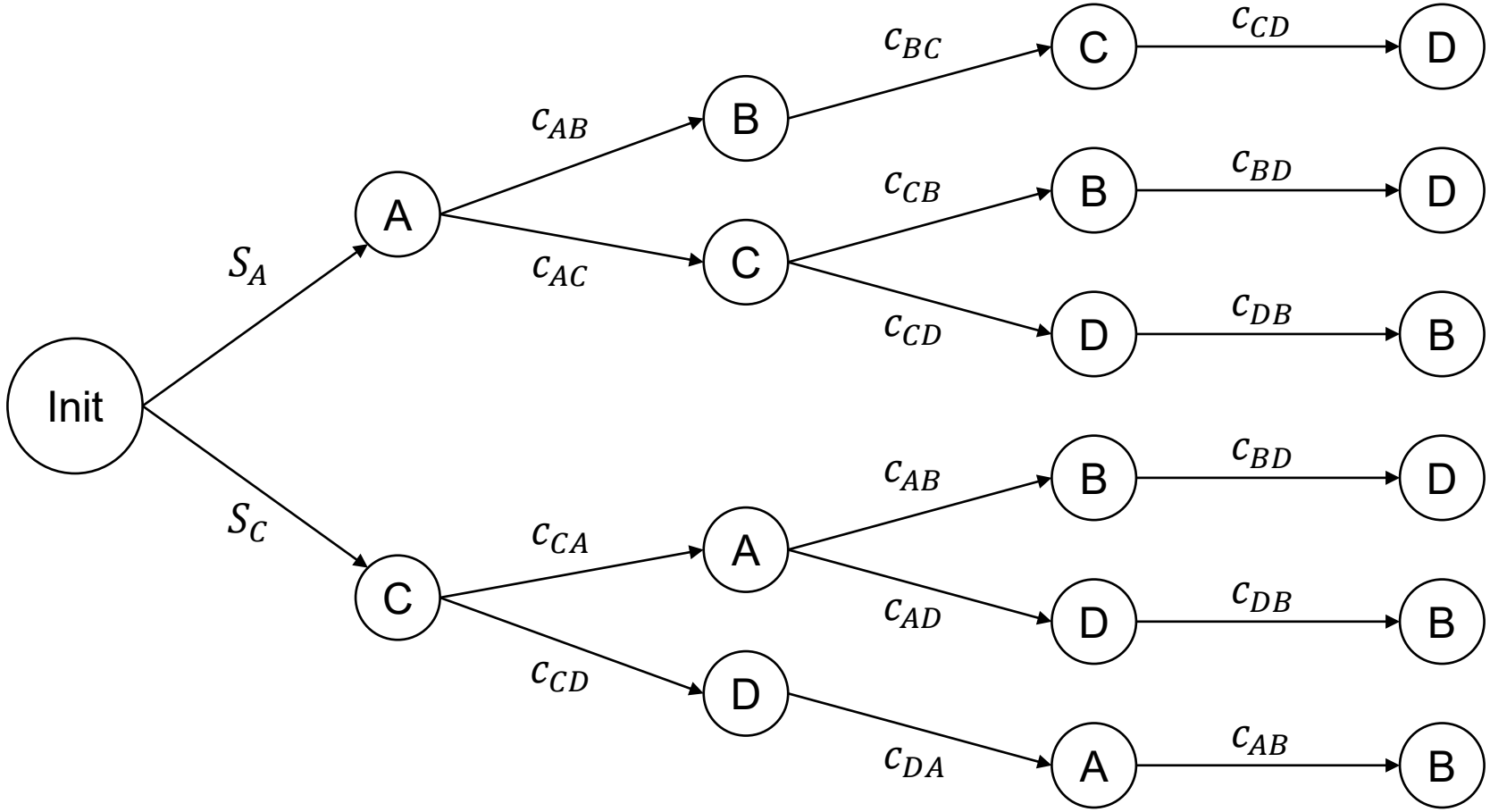
- The idea is to break down the solution (to the combinatorial optimization problems) into components, which can be computed sequentially.

Example: Deterministic Scheduling Problem

- To produce a product
  - Four operations (A, B, C, D) on a machine are needed.
  - Operation B can be performed only after operation A has been performed.
  - Operation D can be performed only after operation C has been performed.
  - Setup cost for passing from operation m to operation n: c_{mn}
  - Initial setup cost for starting with operation A or C: S_A or S_C
  - Goal: Find the optimal scheduling of the operations to minimize the total cost

Source: Bertsekas (2023)

Example: Deterministic Scheduling Problem



Source: Modified from Bertsekas (2023)

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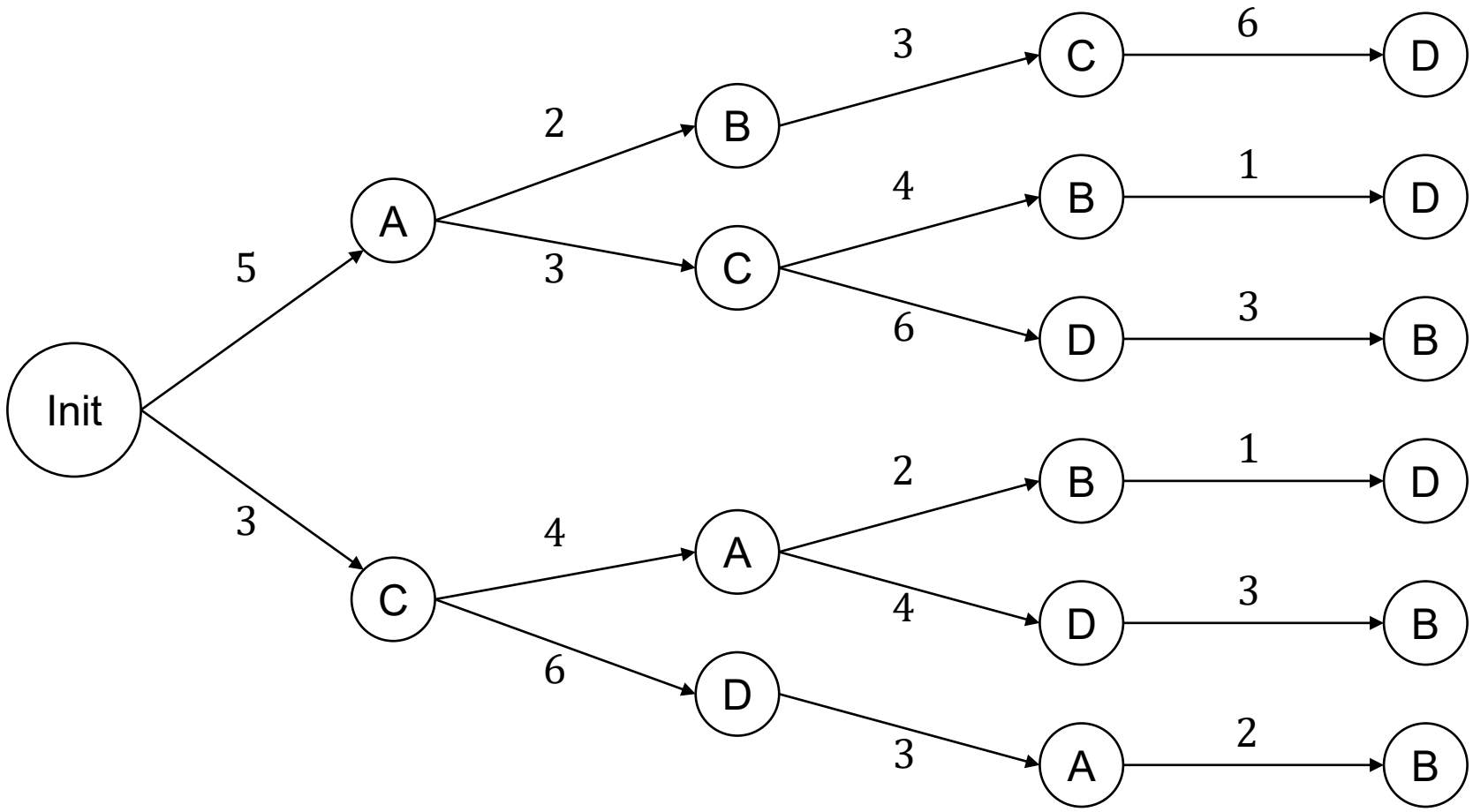


Source: Modified from Bertsekas (2023)

Principle of Optimality

Principle of Optimality

Let $\{x_0^*, \dots, x_{T-1}^*\}$ be an optimal control sequence, which together with S_0 determines the corresponding state sequence $\{S_1, \dots, S_T\}$ via the system equation

$$S_{t+1} = f_t(S_t, x_t),\ t = 0, 1, \dots, T-1.$$

Consider the subproblem (tail subproblem) whereby we start at S_t^* at time t and wish to minimize the "cost-to-go" from time t to time T,

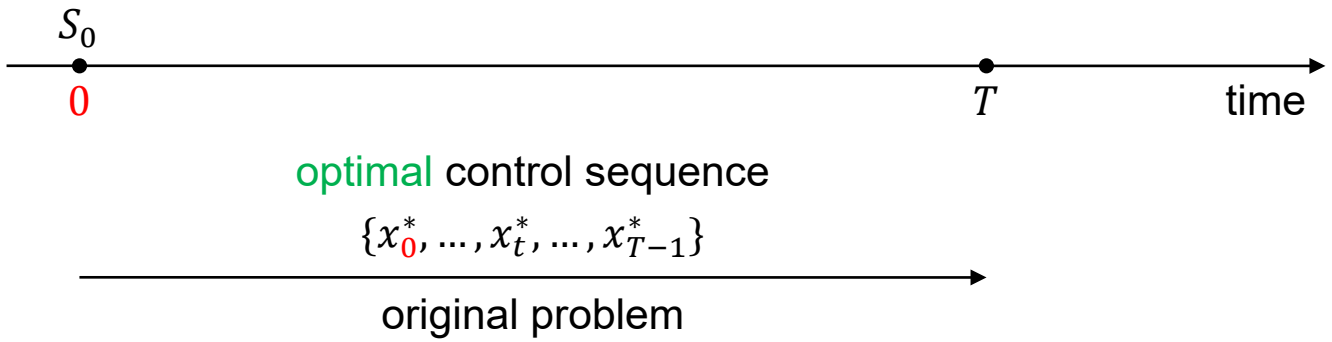
$$c_t(S_t^*, x_t) + \sum_{m=t+1}^{T-1} c_m(S_m, x_m) + c_T(S_T)$$

over $\{x_t, \dots, x_{T-1}\}$ with $x_m \in X_m(S_m)$, $m = t, \dots, T-1$. Then the truncated optimal control sequence $\{x_t^*, \dots, x_{T-1}^*\}$ is optimal for this tail subproblem.

Proof (intuitively)

If $\{x_t^*, \dots, x_{T-1}^*\}$ is not optimal, we would be able to reduce the total cost by switching to an optimal sequence for the tail subproblem once we reach S_t^*, while the preceding control sequence $\{x_0^*, \dots, x_{t-1}^*\}$ remains the same, which does not restrict our future choices.

*Principle of Optimality



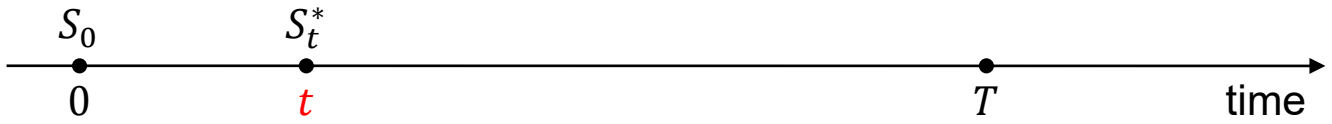
$$\boxed{\min \sum_{t=0}^{T-1} c_t(S_t, x_t) + c_T(S_T)}$$

Source: Modified from Bertsekas (2023)

*Principle of Optimality

$$\min\ c_t(S_t^*, x_t) + \sum_{m=t+1}^{T-1} c_m(S_m, x_m) + c_T(S_T)$$

tail subproblem



optimal control sequence

$$\{x_0^*, \dots, x_t^*, \dots, x_{T-1}^*\}$$

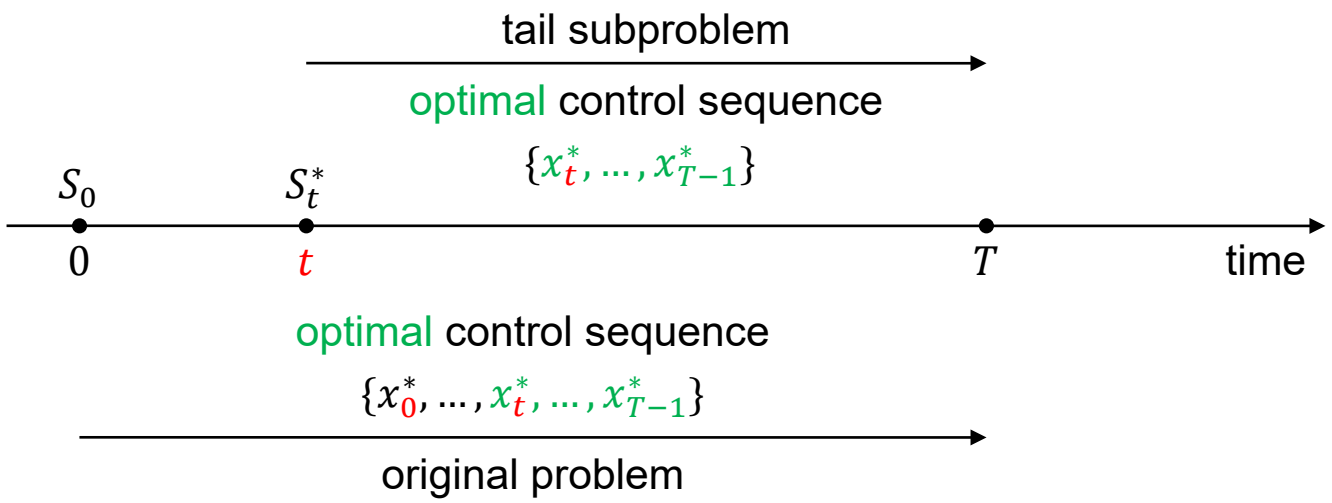
original problem

$$\min\ \sum_{t=0}^{T-1} c_t(S_t, x_t) + c_T(S_T)$$

Source: Modified from Bertsekas (2023)

*Principle of Optimality

$$\min\ c_t(S_t^*, x_t) + \sum_{m=t+1}^{T-1} c_m(S_m, x_m) + c_T(S_T)$$



$$\min\ \sum_{t=0}^{T-1} c_t(S_t, x_t) + c_T(S_T)$$

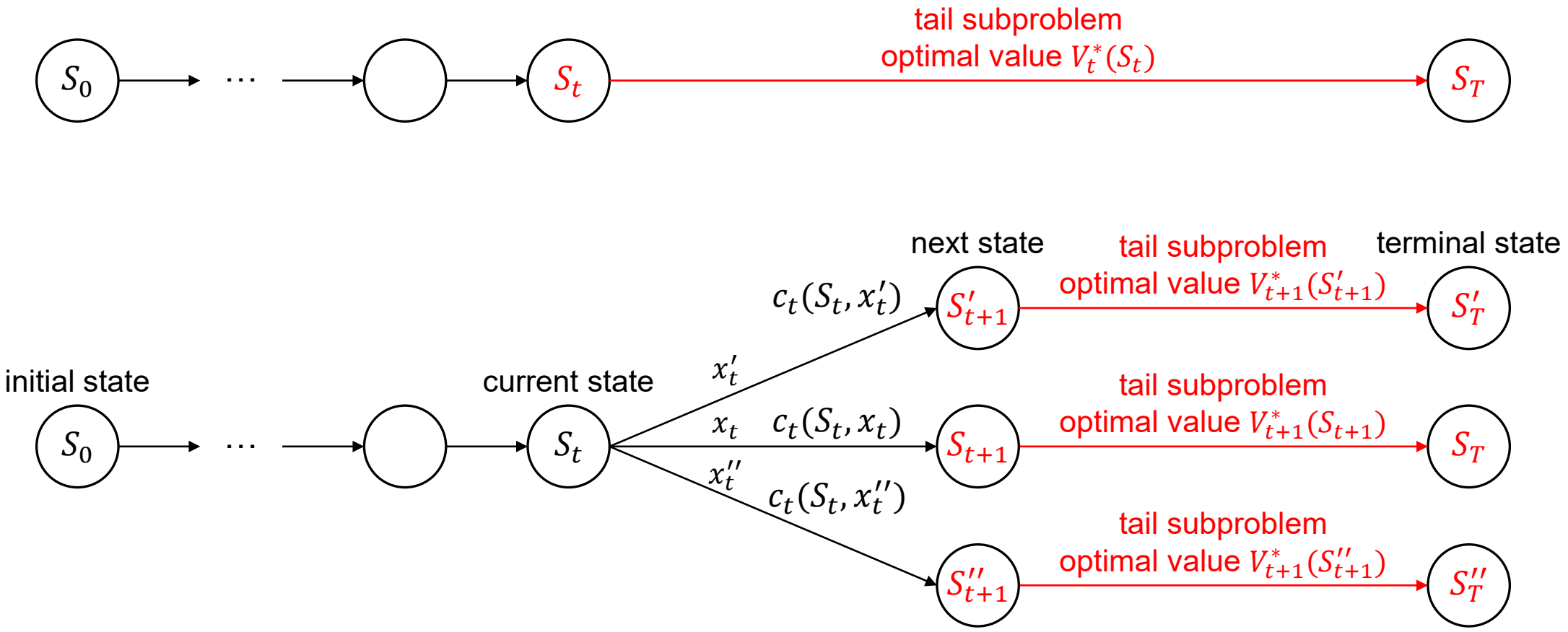
Source: Modified from Bertsekas (2023)

Dynamic Programming (DP) Algorithm

*DP Algorithm

Source: Modified from Bertsekas (2023)

*DP Algorithm



Source: Modified from Bertsekas (2023)

DP Algorithm for Deterministic Finite Horizon Problems

DP algorithm

Start with

$$V_T^*(S_T) = c_T(S_T),\ \forall S_T.$$

Sequentially, going backward, for $t = T-1, \dots, 0$, let

$$V_t^*(S_t) = \min_{x_t \in X_t(S_t)} [c_t(S_t, x_t) + V_{t+1}^*(f_t(S_t, x_t))],\ \forall S_t.$$

Construction of optimal control sequence

Set

$$x_0^* \in \arg\min_{x_0 \in X_0(S_0)} [c_0(S_0, x_0) + V_1^*(f_0(S_0, x_0))],$$

and

$$S_1^* = f_0(S_0, x_0^*).$$

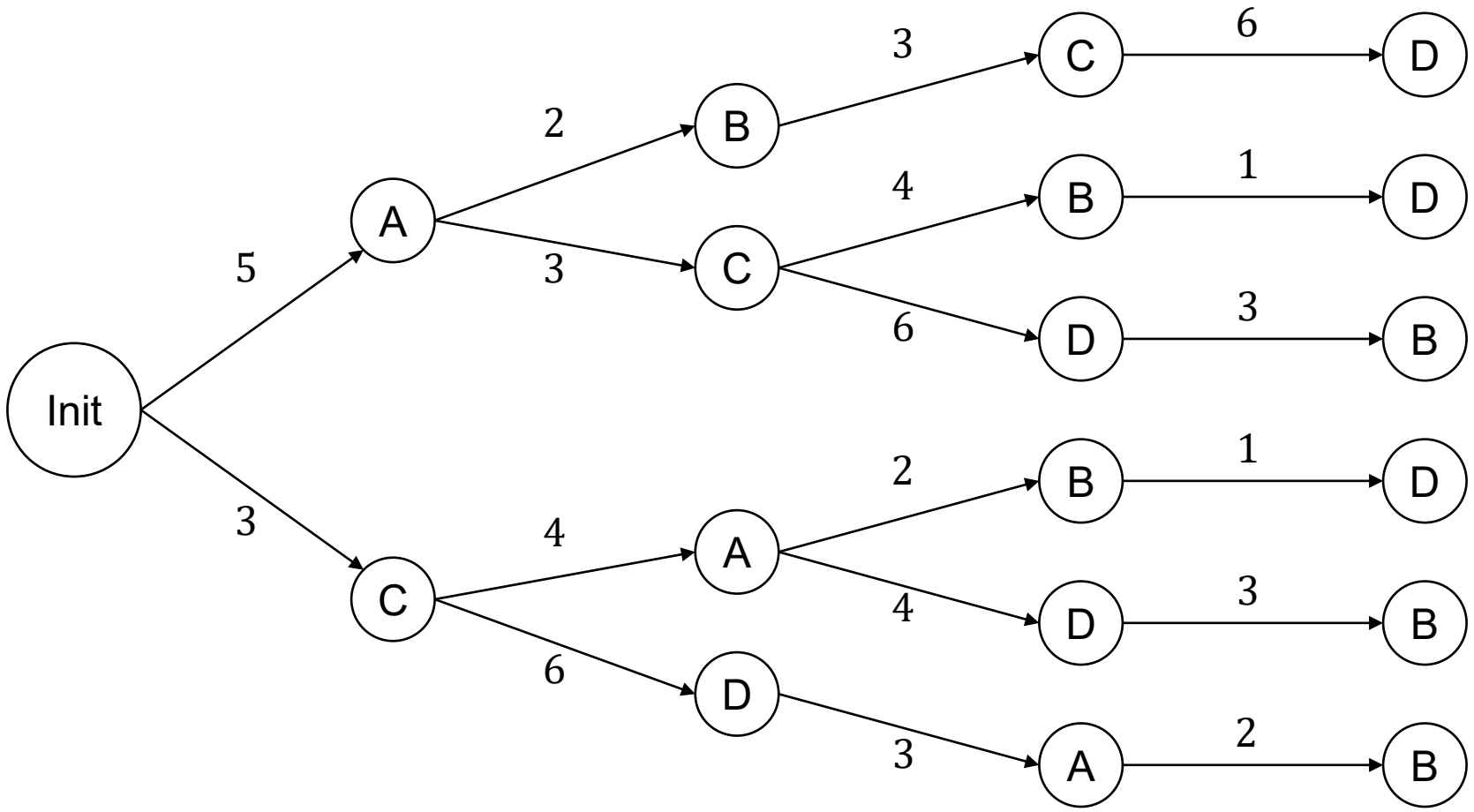
Sequentially, going forward, for $t = 1, \dots, T-1$, set

$$x_t^* \in \arg\min_{x_t \in X_t(S_t)} [c_t(S_t, x_t) + V_{t+1}^*(f_t(S_t, x_t))],$$

And

$$S_{t+1}^* = f_t(S_t, x_t^*).$$

Example: Deterministic Scheduling Problem



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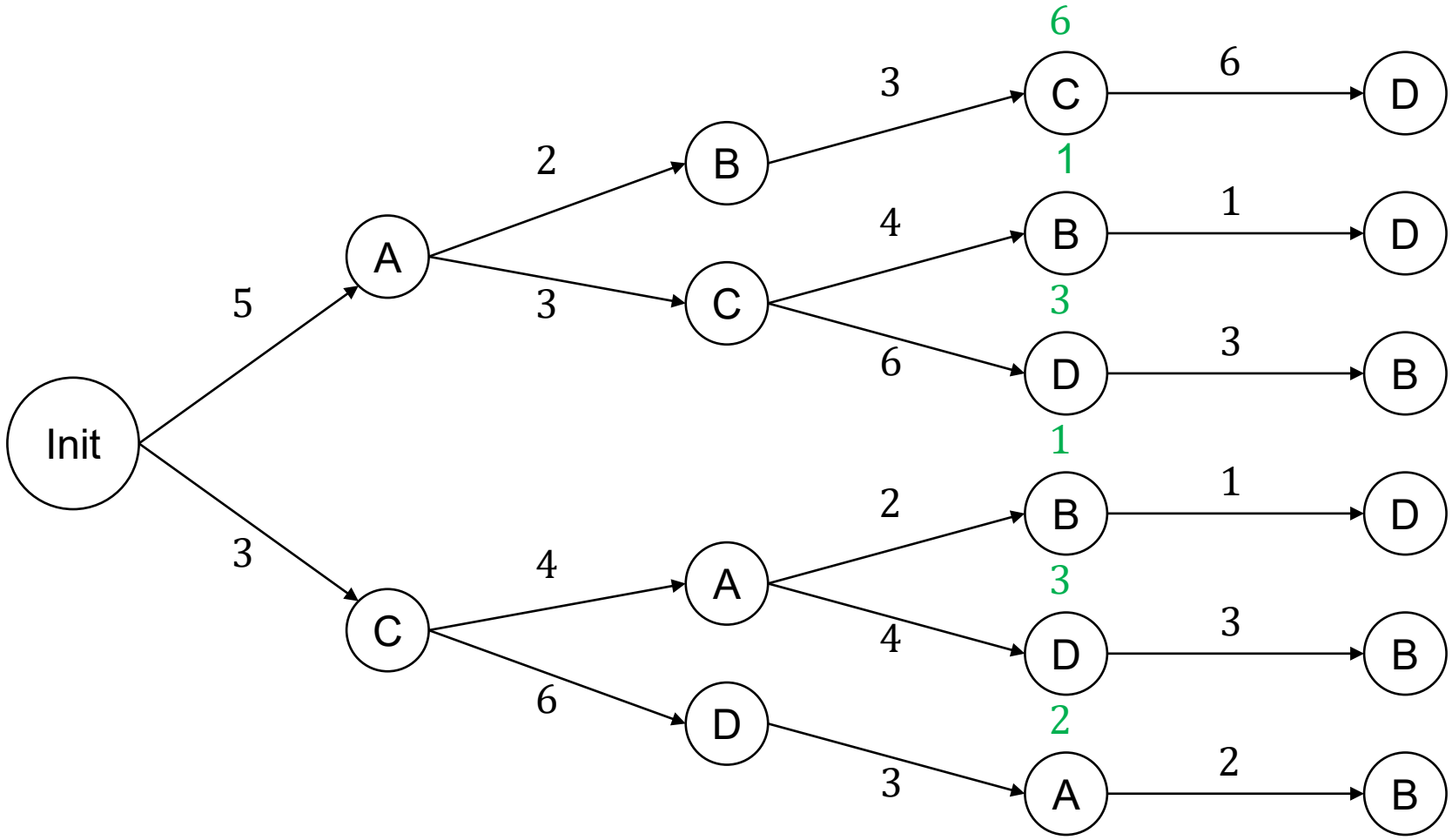

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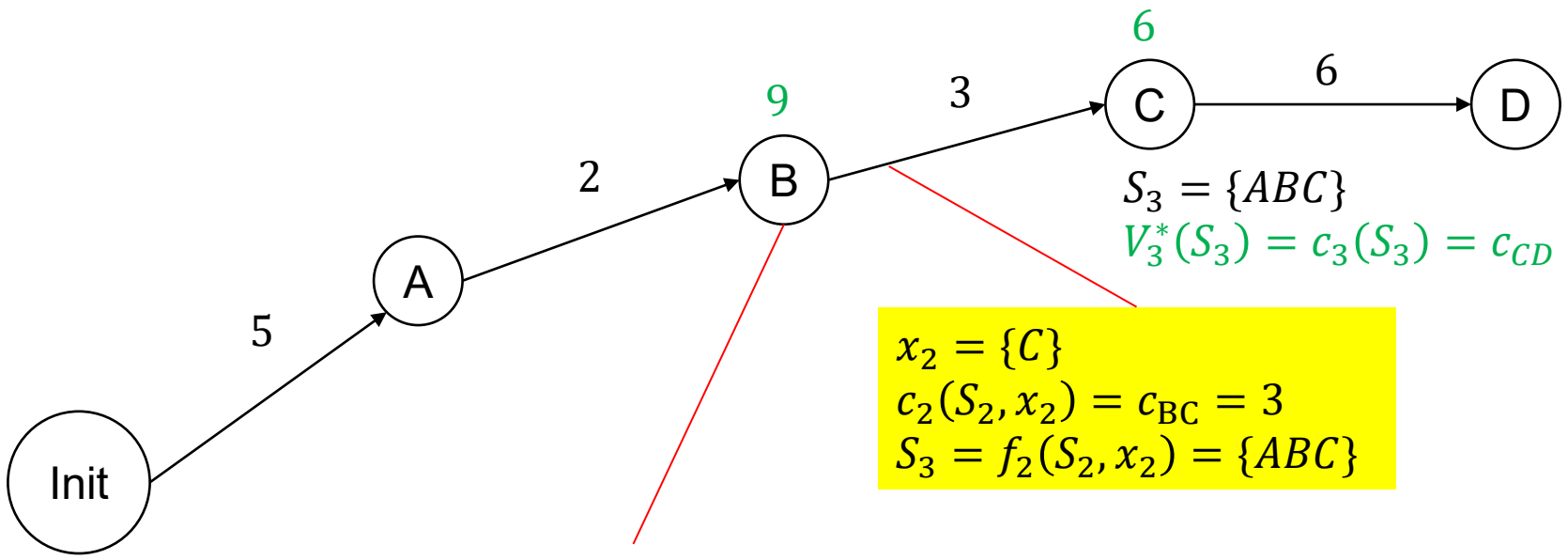


$S_3 = \{ABC\}$

$V_3^*(S_3) = c_3(S_3) = c_{CD} = 6$

$x_2 = \{C\}$

$c_2(S_2, x_2) = c_{\mathrm{BC}} = 3$

$S_3 = f_2(S_2, x_2) = \{ABC\}$

$S_2 = \{AB\}$

$X_2(S_2) = \{C\}$

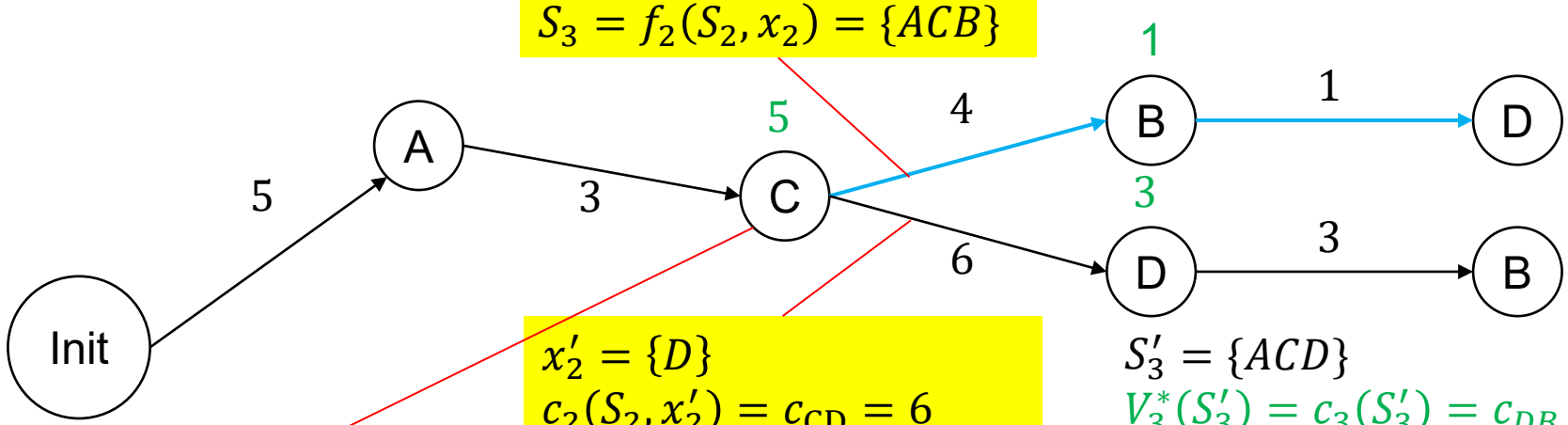
$V_2^*(S_2) = c_2(S_2, x_2) + V_3^*(S_3) = 9$

Source: Modified from Bertsekas (2023)

*Example: Deterministic Scheduling Problem

$x_2 = \{B\}$
$c_2(S_2, x_2) = c_{\mathrm{CB}} = 4$
$S_3 = f_2(S_2, x_2) = \{ACB\}$

$S_3 = \{ACB\}$
$V_3^*(S_3) = c_3(S_3) = c_{BD} = 1$



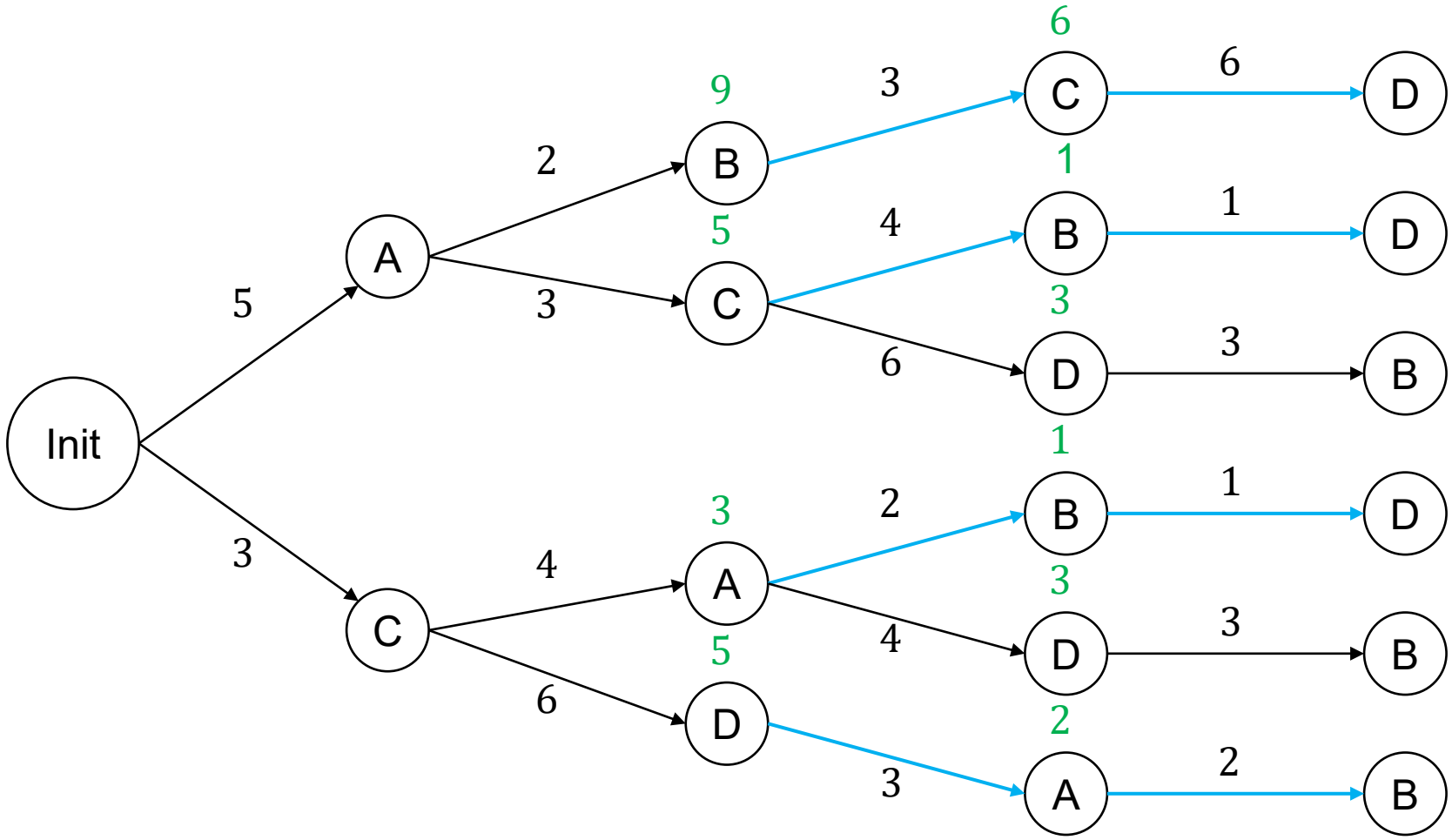
$x_2' = \{D\}$
$c_2(S_2, x_2') = c_{\mathrm{CD}} = 6$
$S_3' = f_2(S_2, x_2') = \{ACD\}$

$S_3' = \{ACD\}$
$V_3^*(S_3') = c_3(S_3') = c_{DB} = 3$

$S_2 = \{AC\}$
$X_2(S_2) = \{B, D\}$
$V_2^*(S_2) = \min_{x_2 \in X_2(S_2)} \left[c_2(S_2, x_2) + V_3^*\left(f_2(S_2, x_2)\right)\right] = \min\{4 + 1, 6 + 3\} = 5$

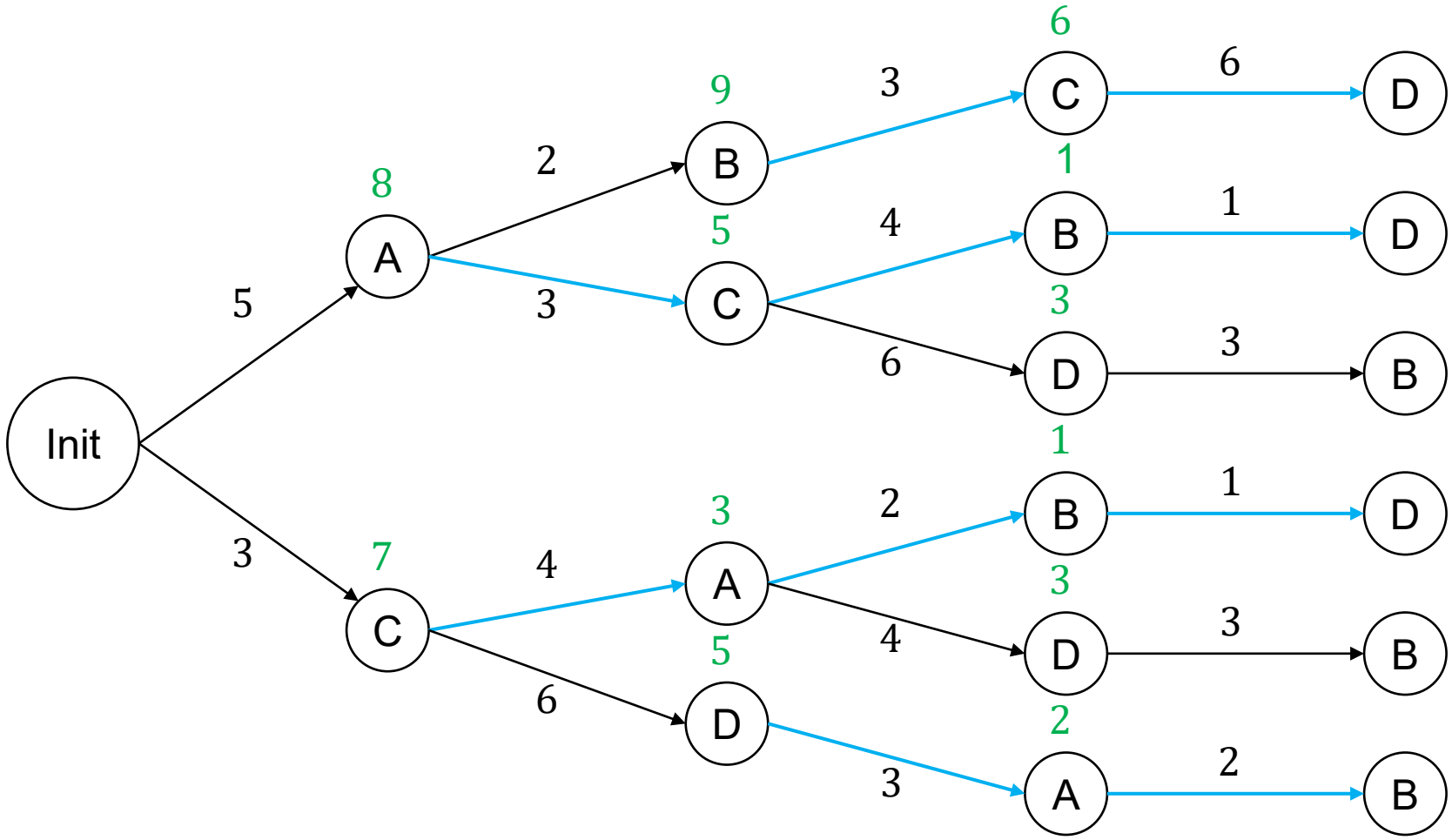
Source: Modified from Bertsekas (2023)

*Example: Deterministic Scheduling Problem



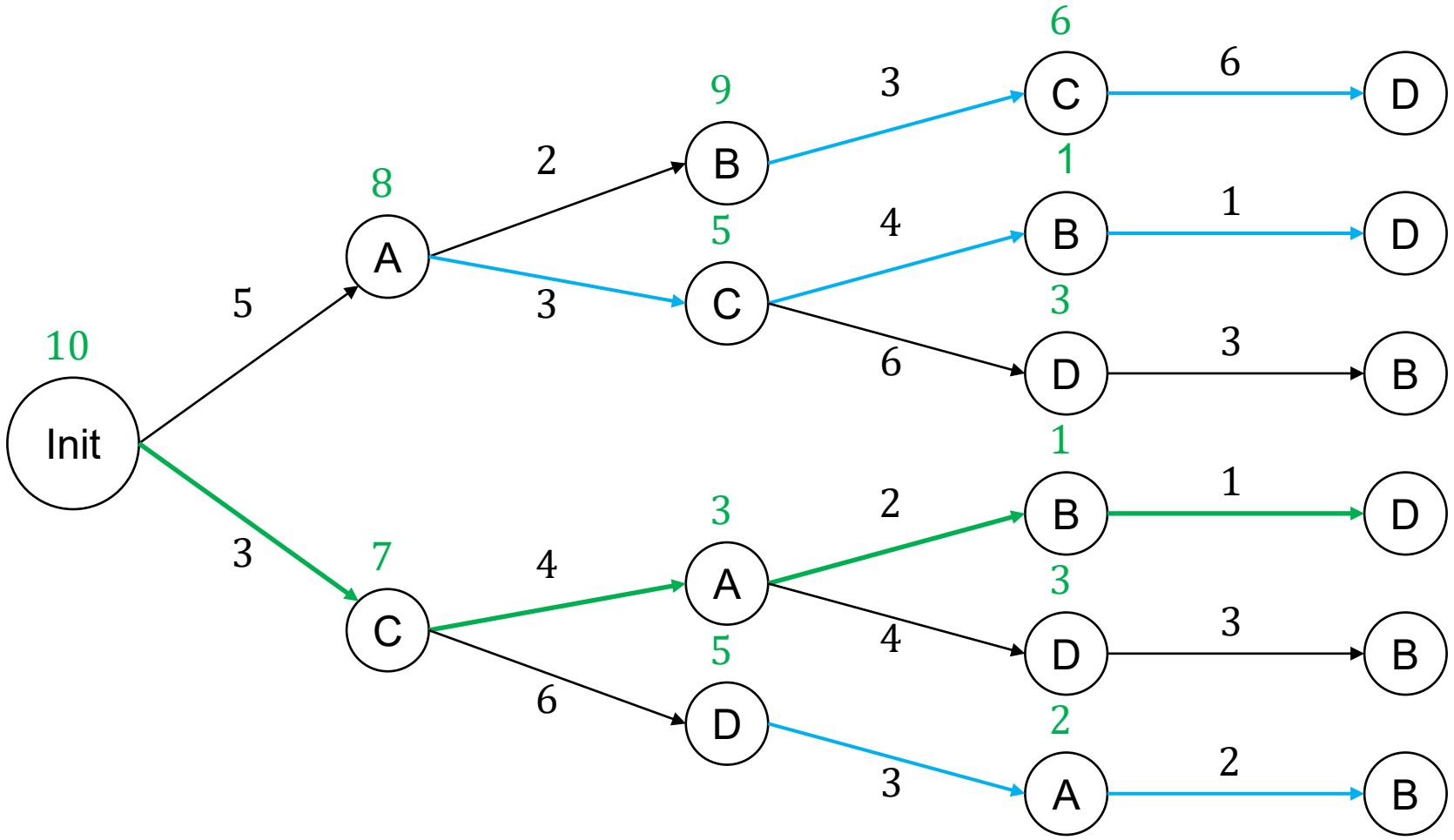
Source: Modified from Bertsekas (2023)

*Example: Deterministic Scheduling Problem



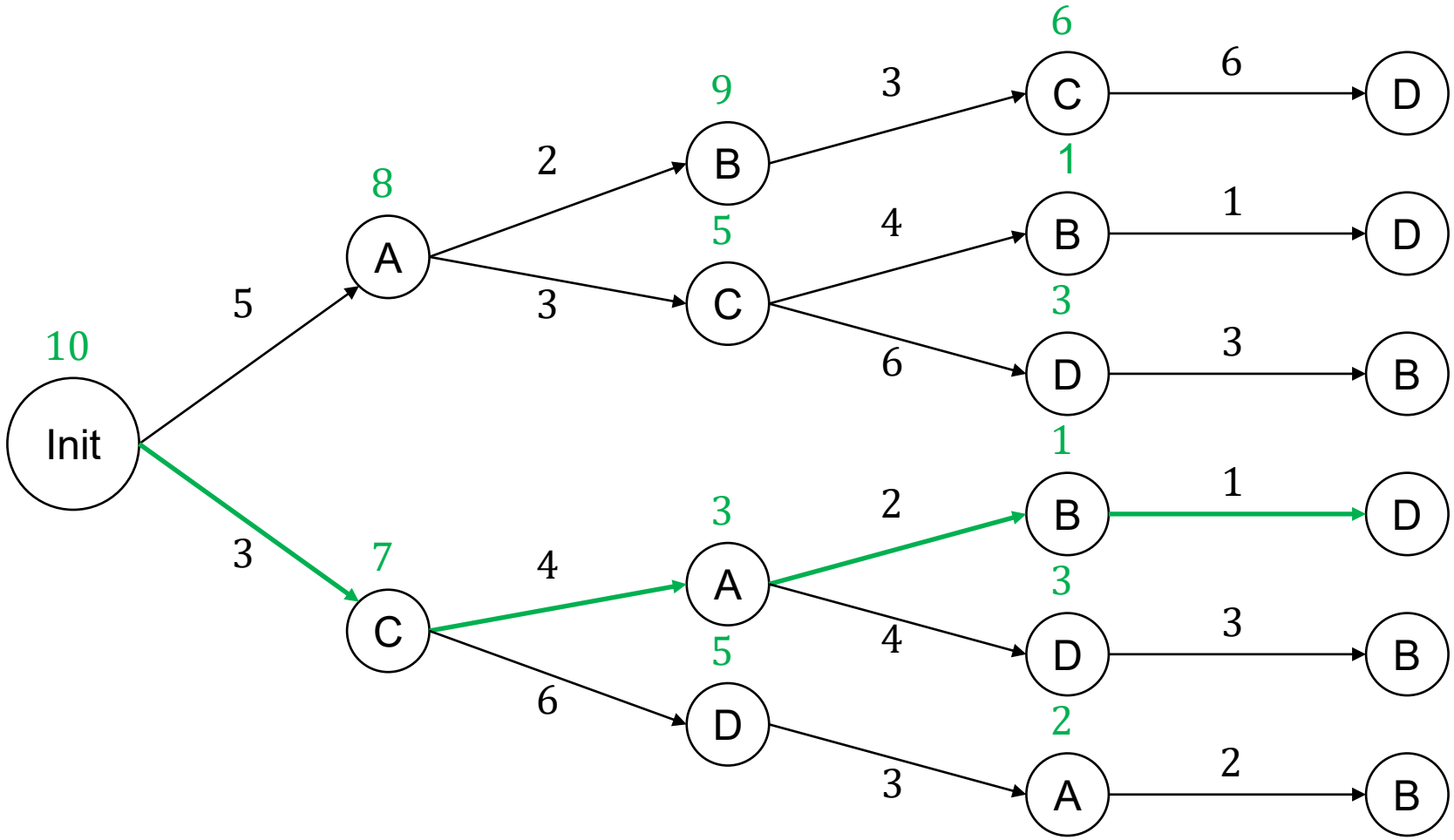
Source: Modified from Bertsekas (2023)

*Example: Deterministic Scheduling Problem



Source: Modified from Bertsekas (2023)

*Example: Deterministic Scheduling Problem

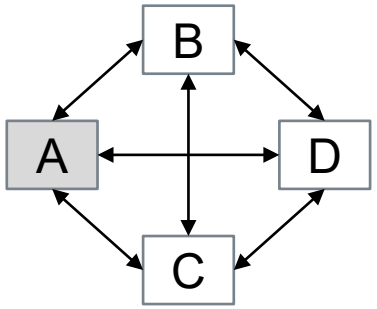


Source: Modified from Bertsekas (2023)

Example: Traveling Salesman Problem

initial state S_0

A



A

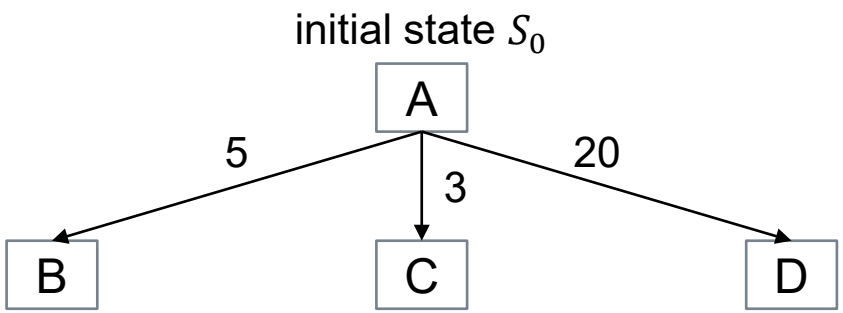
terminal state S_T

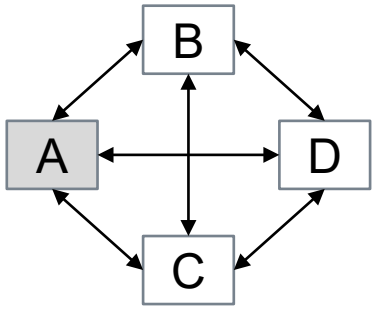
to

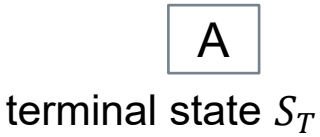
from	A	B	C	D
A		5	3	20
B	20		1	4
C	1	20		3
D	20	4	3	

Source: Modified from Bertsekas (2023)

*Example: Traveling Salesman Problem







to

from	A	B	C	D
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Source: Modified from Bertsekas (2023)

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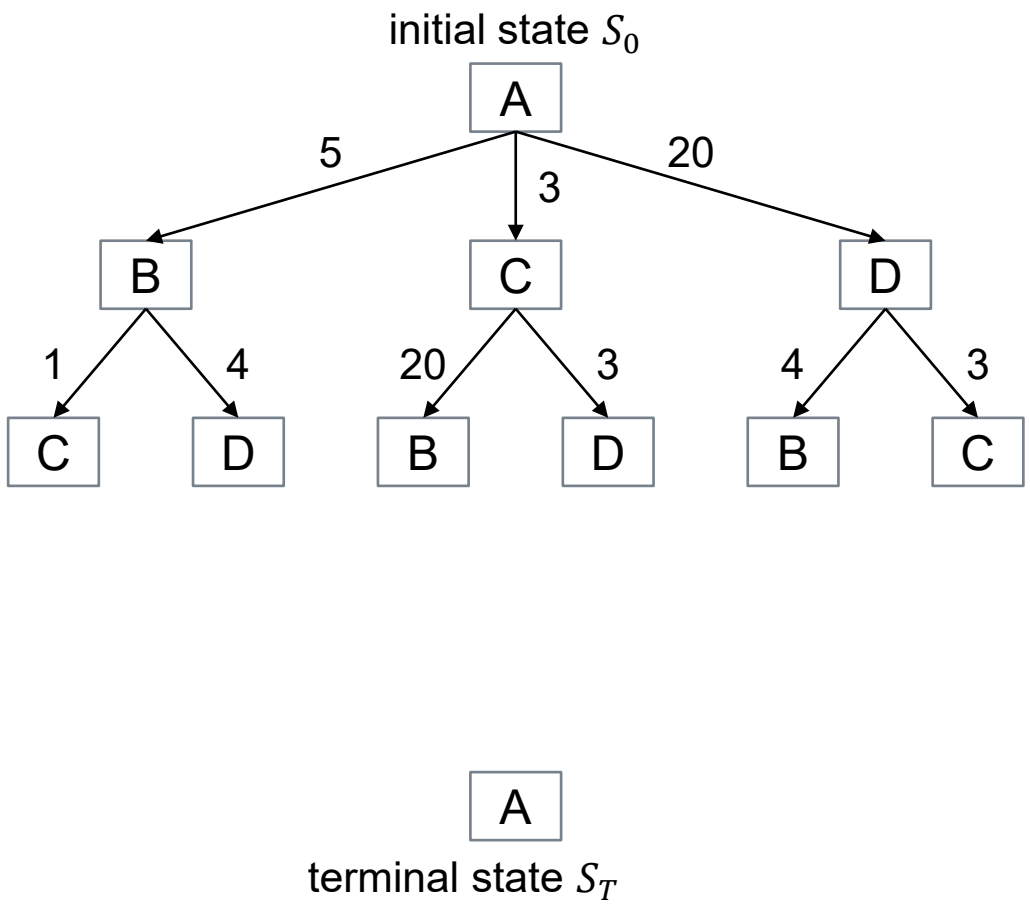

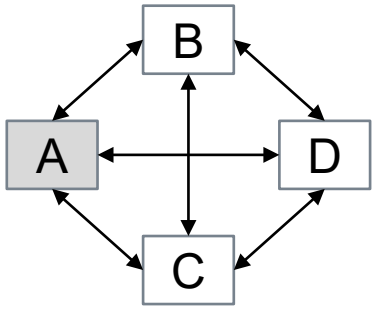

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Source: Modified from Bertsekas (2023)

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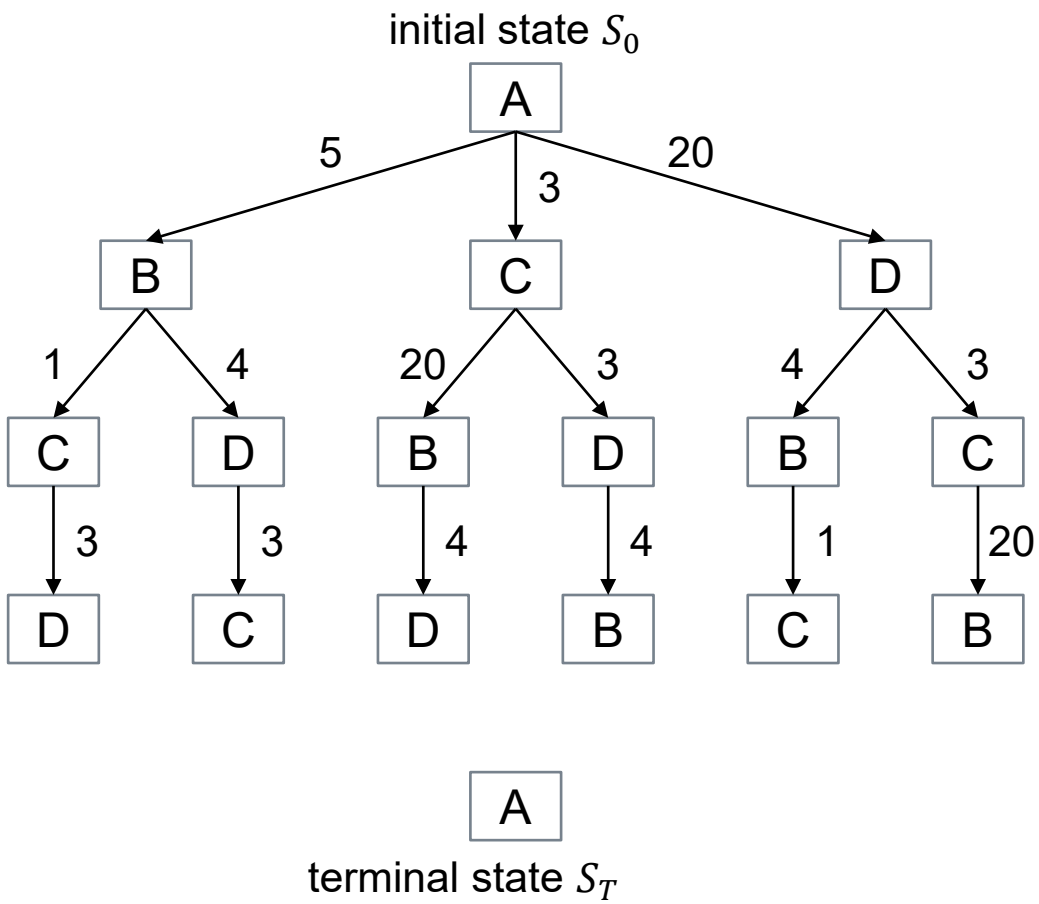

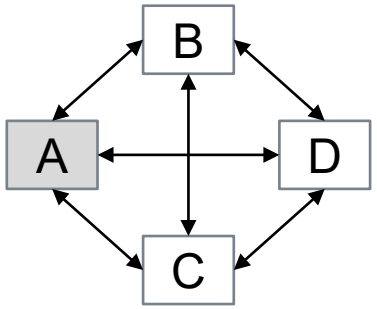

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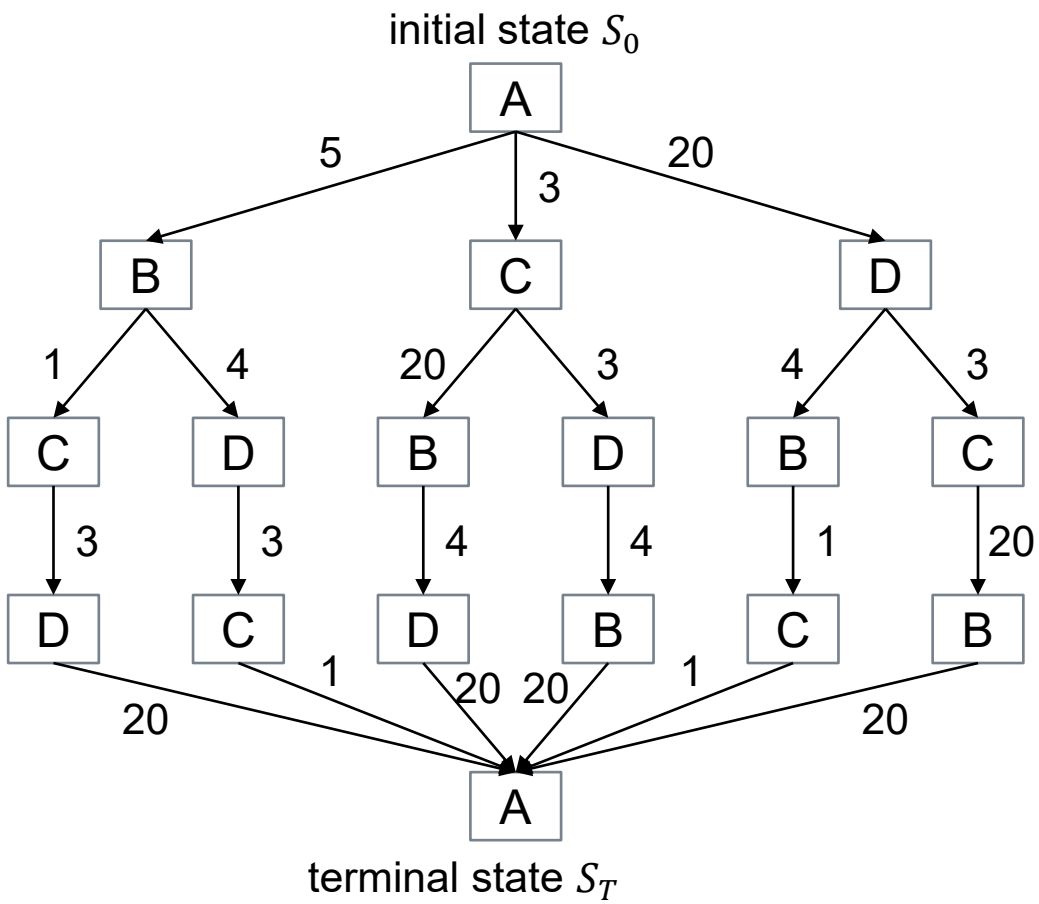

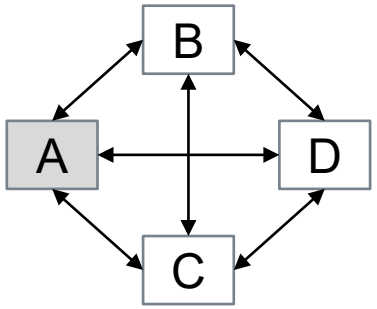

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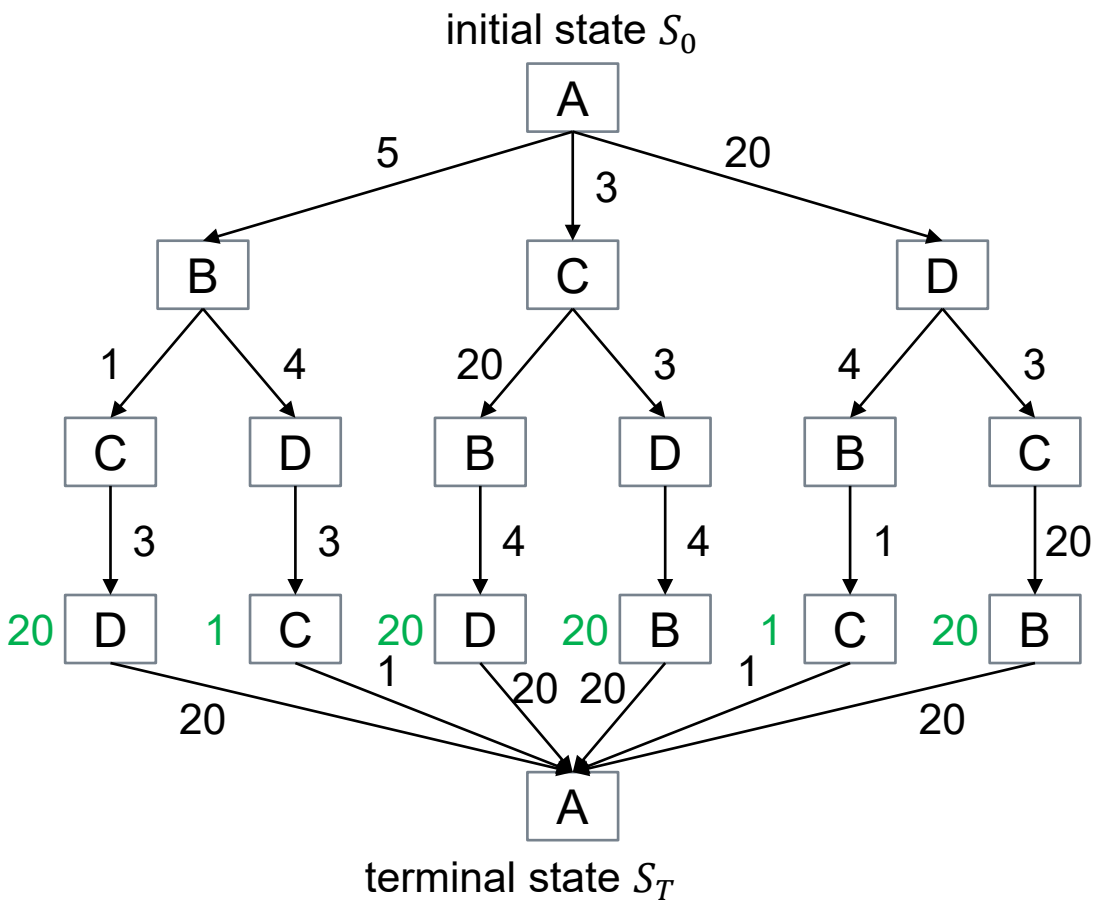

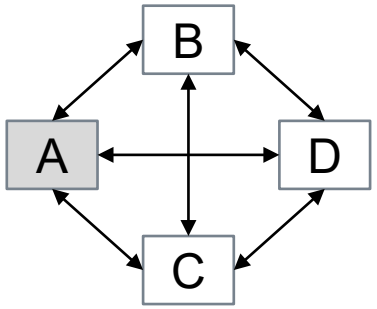

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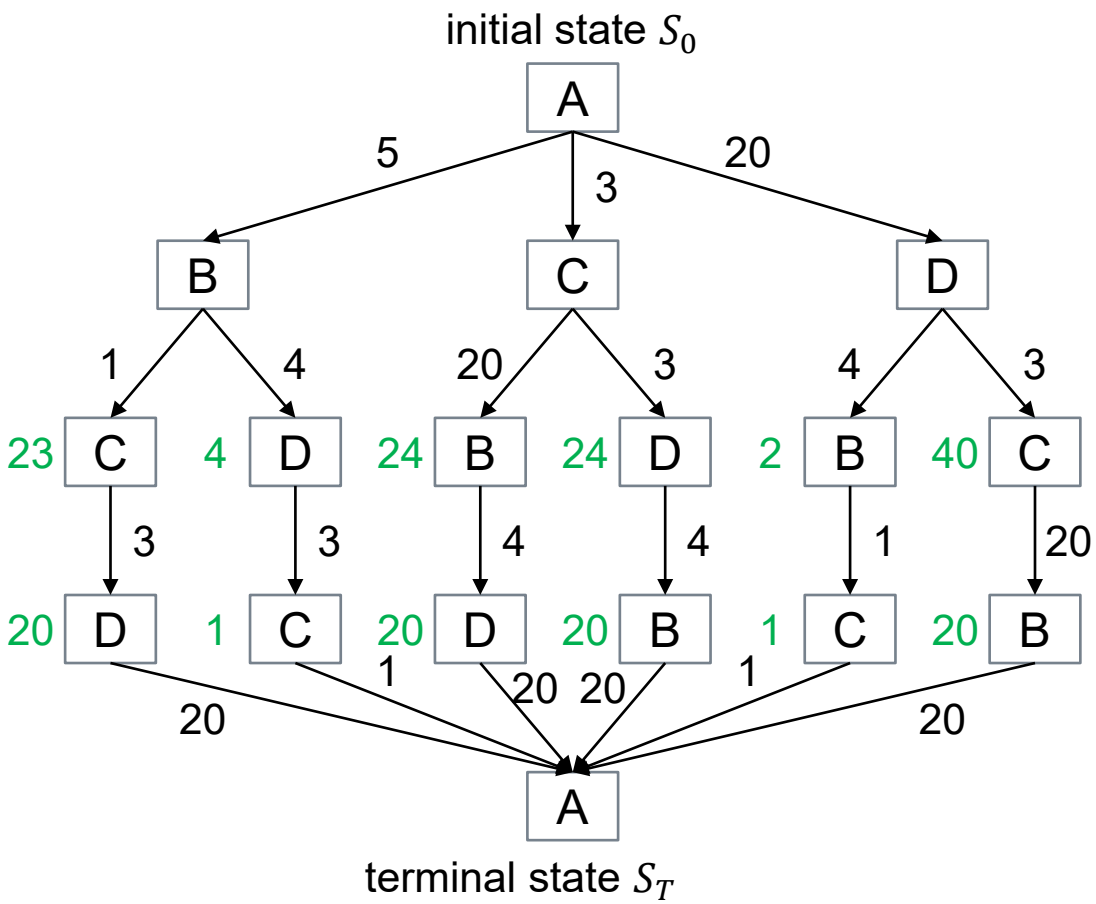

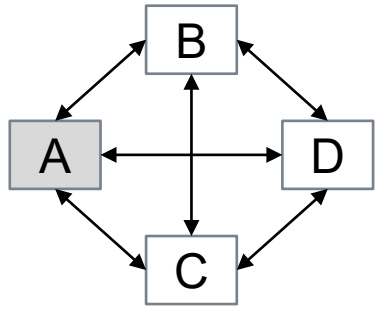

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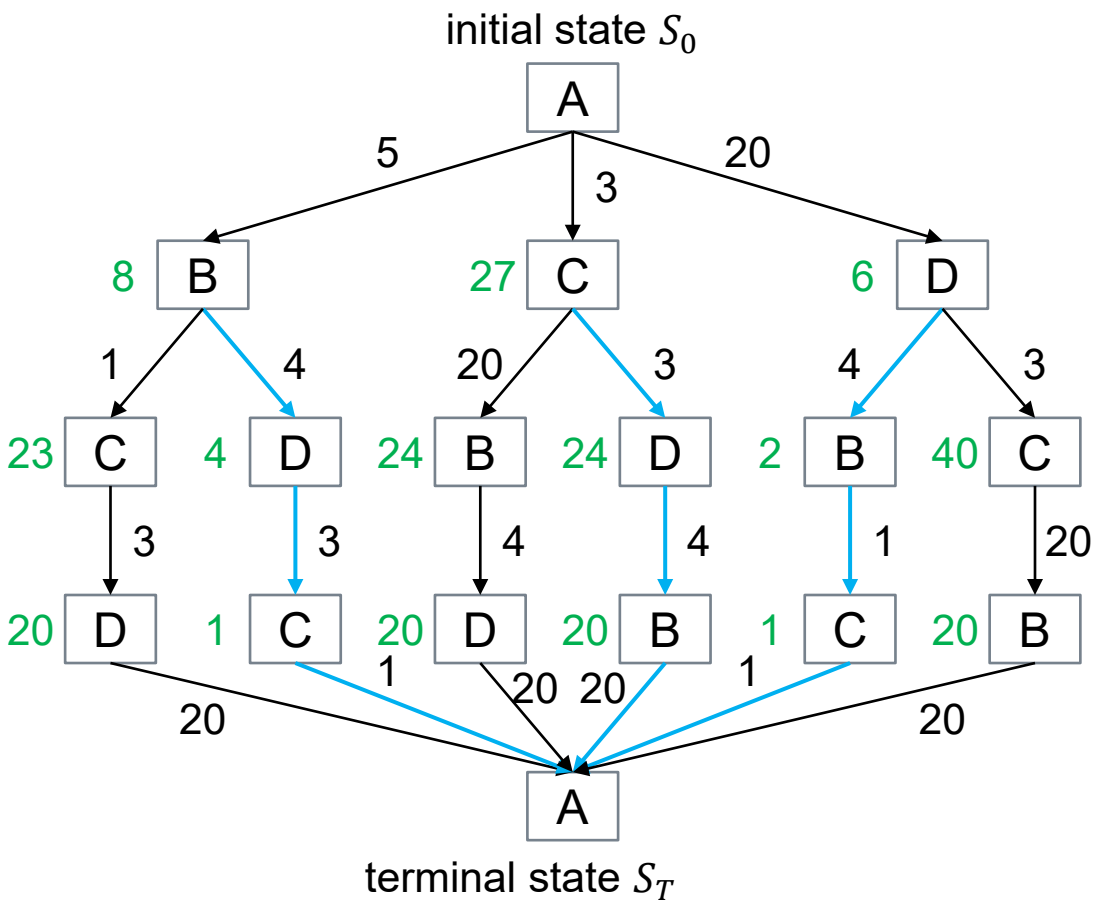

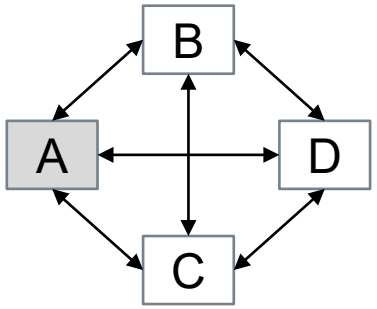

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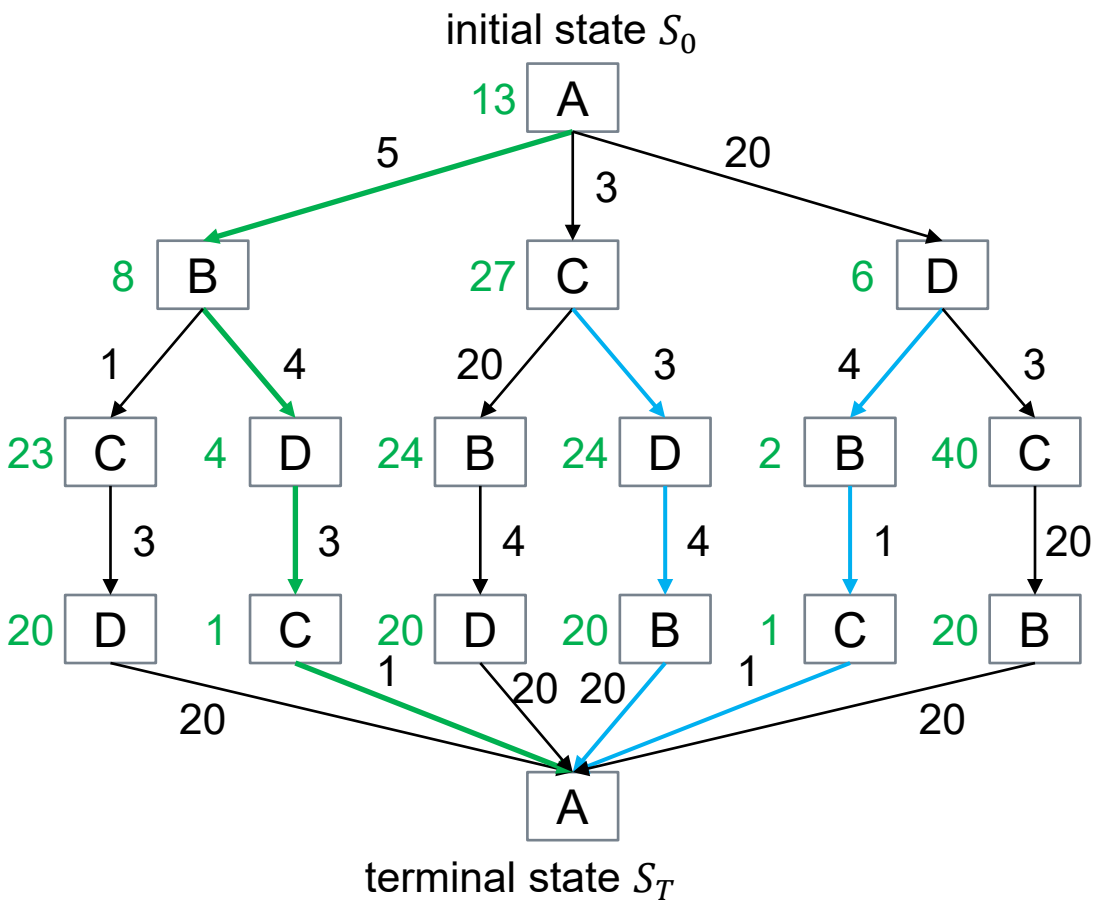

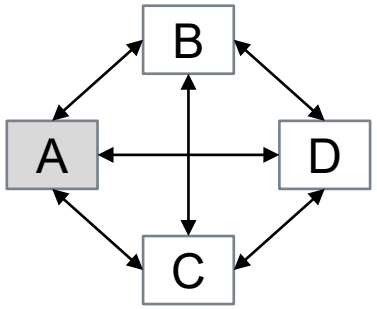

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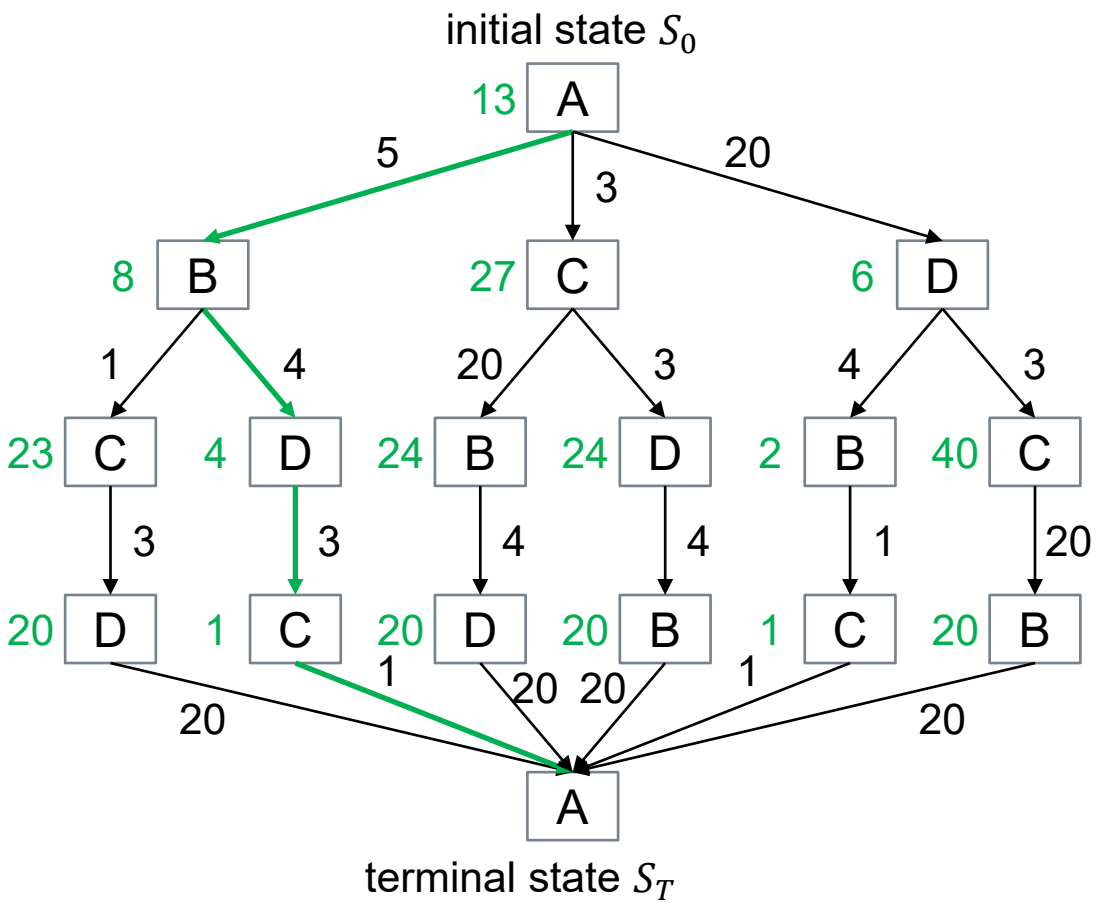
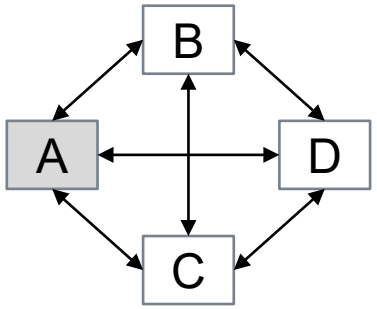
Source: Modified from Bertsekas (2023)

*Example: Traveling Salesman Problem





to

from	A	B	C	D
A		5	3	20
B	20		1	4
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D	20	4	3	

Source: Modified from Bertsekas (2023)

Rollout Algorithm

Deterministic Finite State Finite Horizon Problem (Recall)

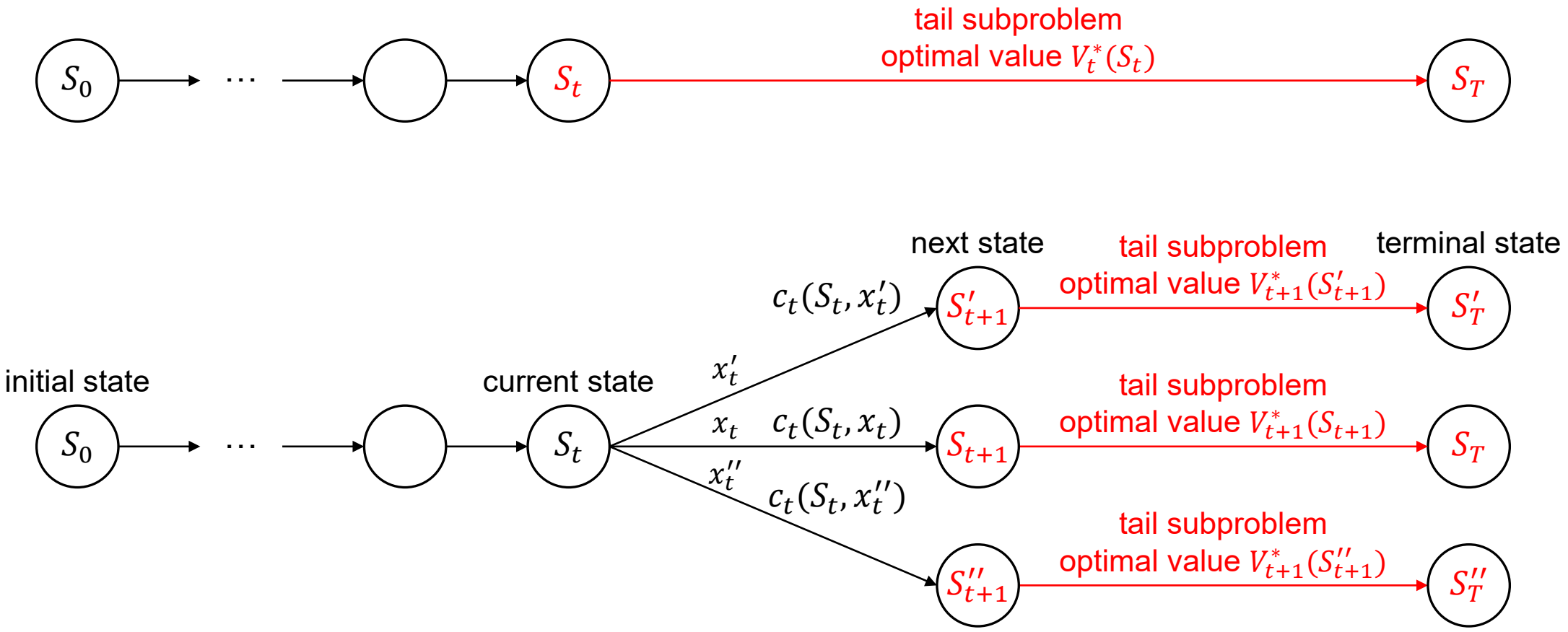
- A deterministic finite state finite horizon problem is equivalent to finding a shortest path from the initial nodes of the (acyclic) graph to the terminal node.

- Generally, combinatorial optimization problems can be formulated as deterministic finite state finite horizon optimal control problem.

- The idea is to break down the solution (to the combinatorial optimization problems) into components, which can be computed sequentially.

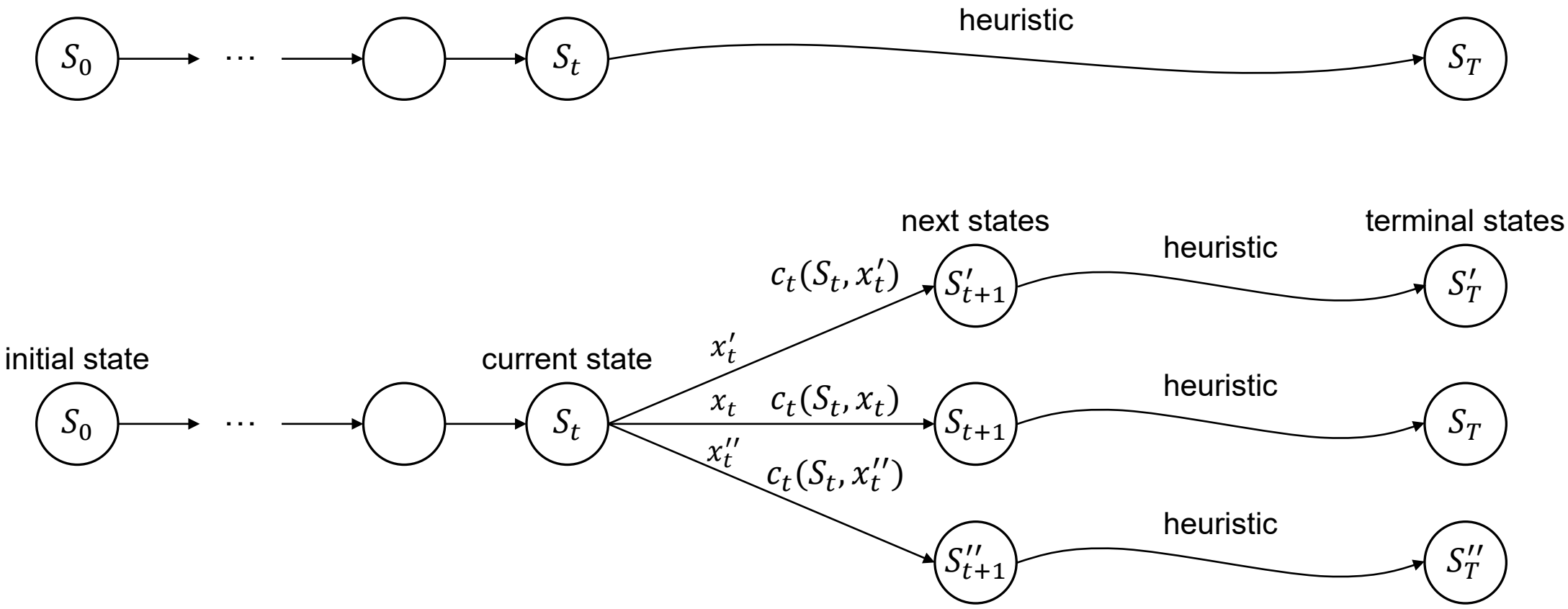
- Exponential growth!!!

*DP Algorithm (Recall)



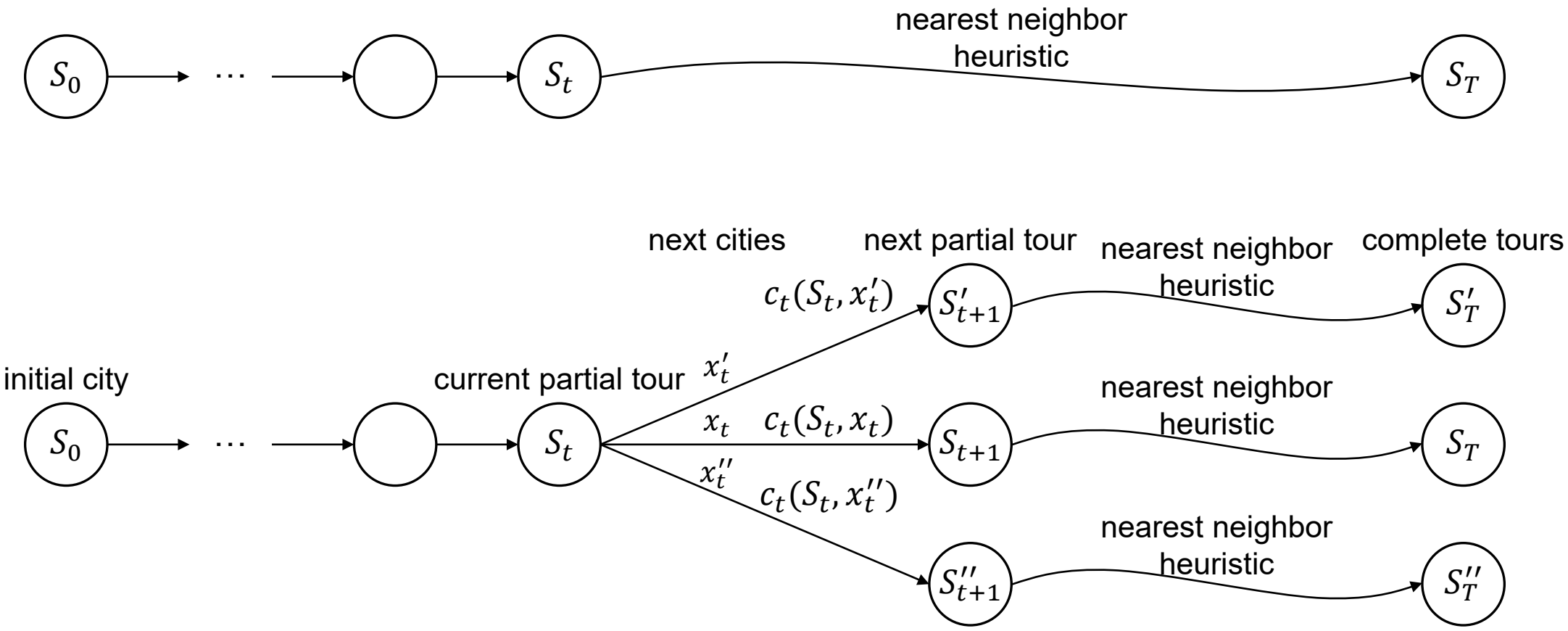
Source: Modified from Bertsekas (2023)

Rollout with One-step Lookahead



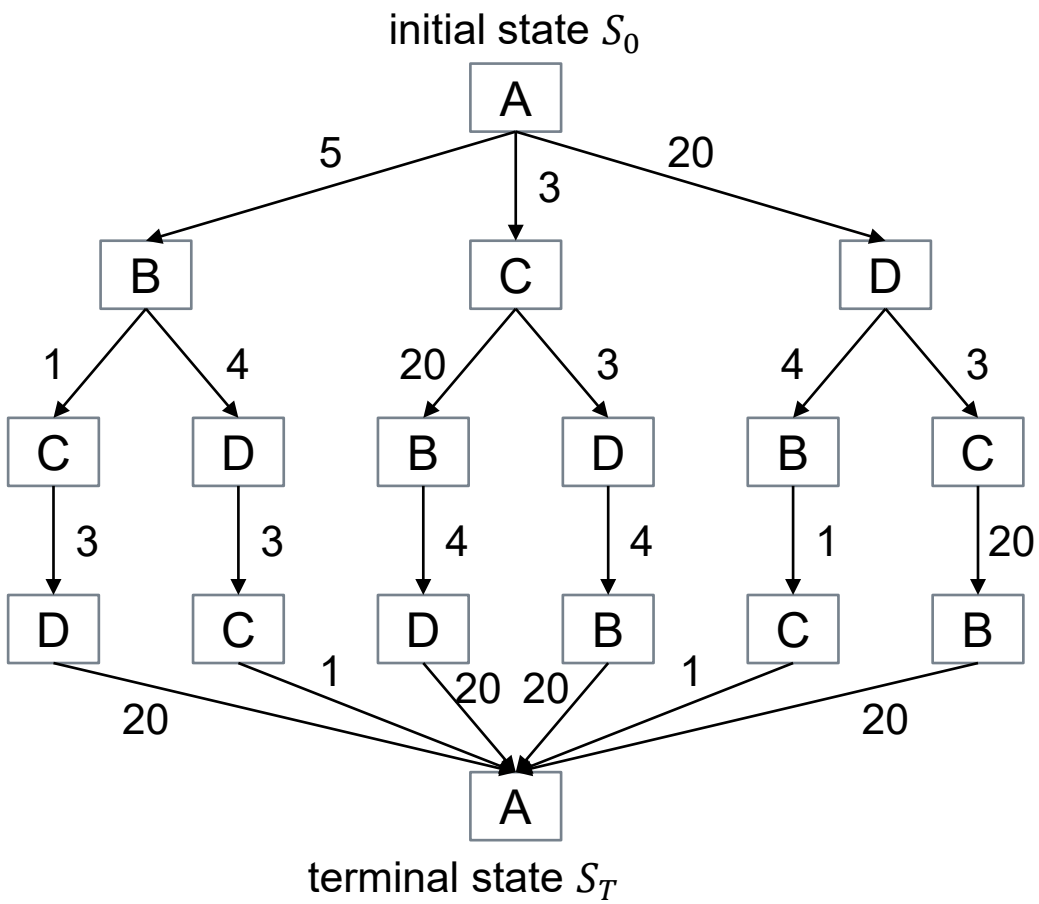
Source: Modified from Bertsekas (2023)

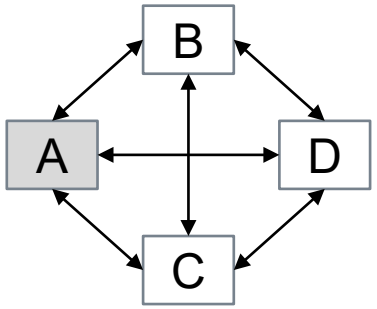
Rollout with One-step Lookahead (TSP)



Source: Modified from Bertsekas (2023)

*Example: Traveling Salesman Problem



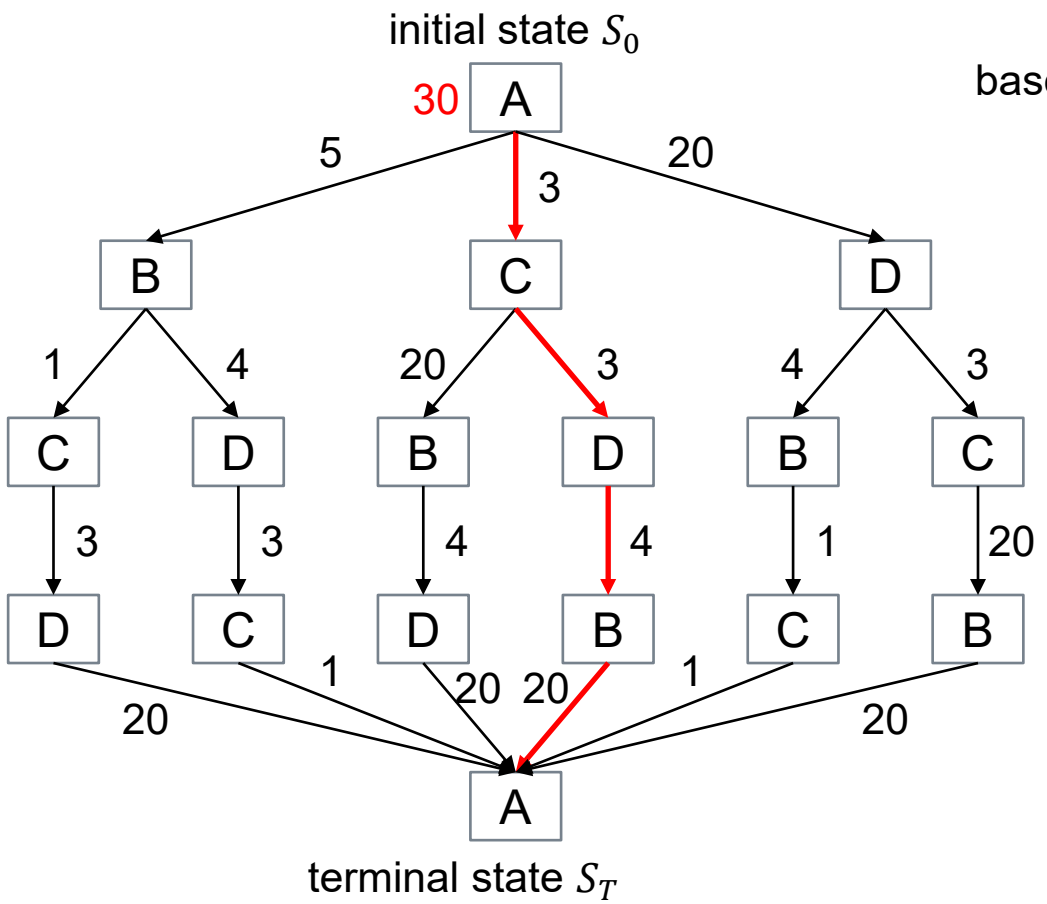


to

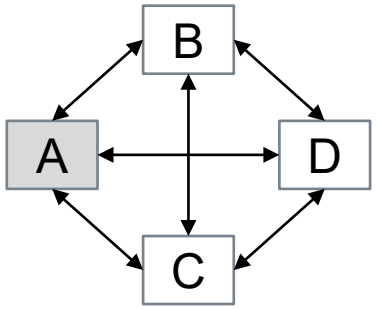
from	A	B	C	D
A		5	3	20
B	20		1	4
C	1	20		3
D	20	4	3	

Source: Modified from Bertsekas (2023)

*Example: Traveling Salesman Problem (Nearest Neighborhood)



base heuristic
T_0

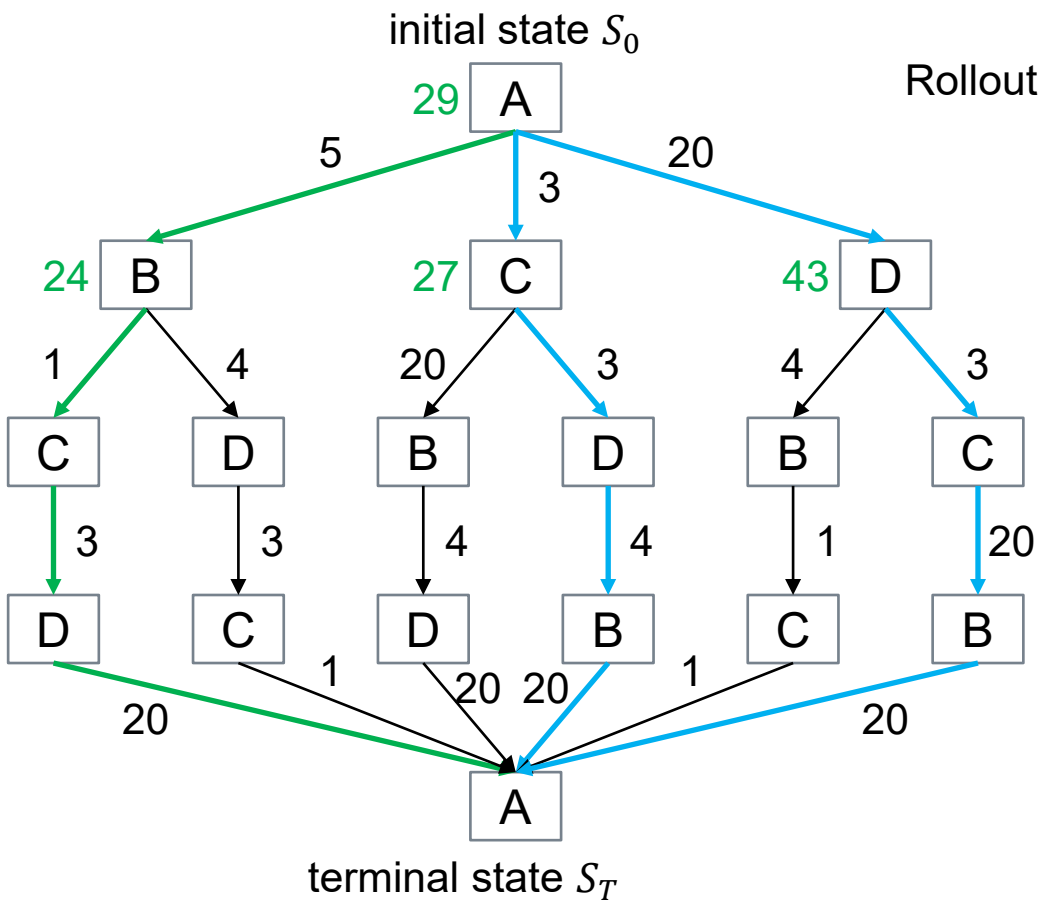


to

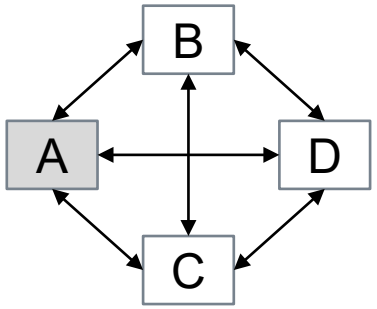
from	A	B	C	D
A		5	3	20
B	20		1	4
C	1	20		3
D	20	4	3	

Source: Modified from Bertsekas (2023)

*Example: Traveling Salesman Problem (Rollout at City A)



Rollout algorithm at A
T_1

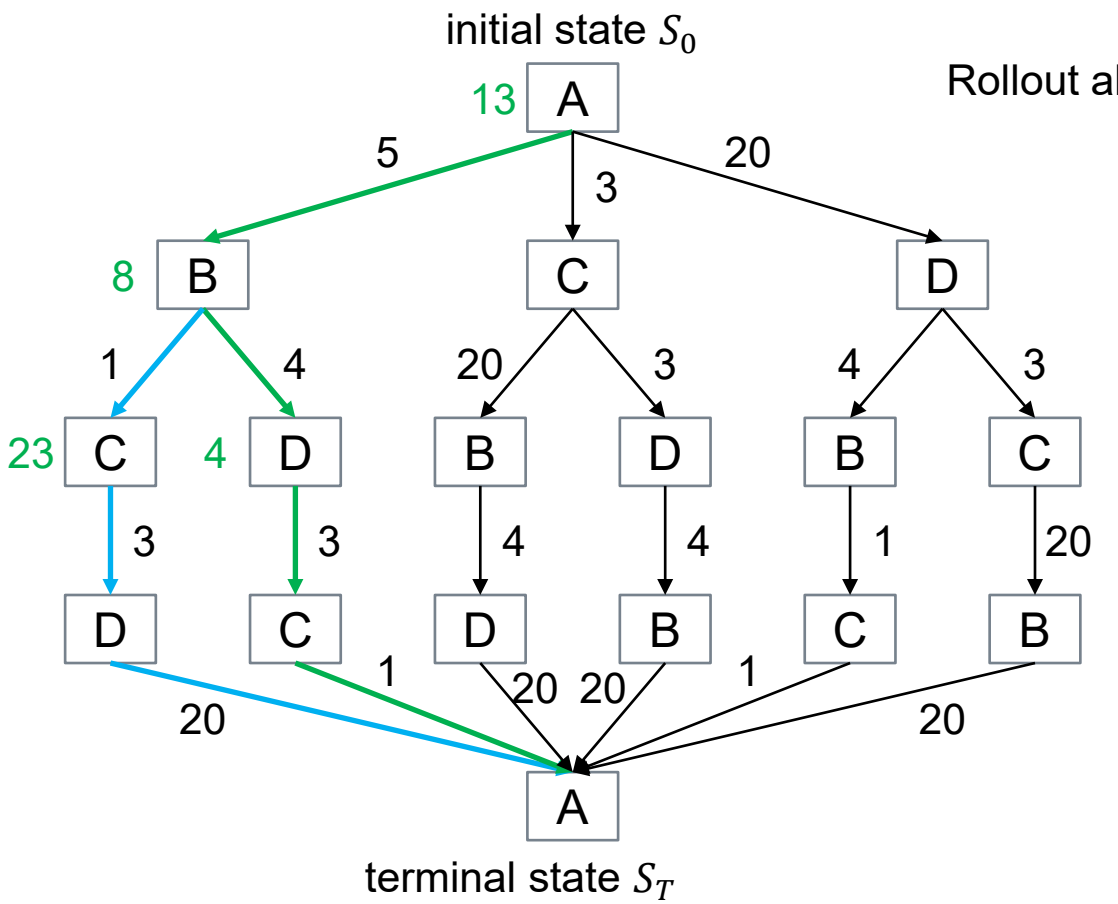


to

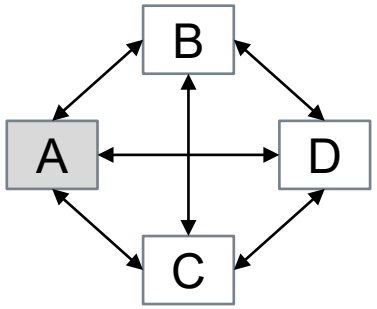
from	A	B	C	D
A		5	3	20
B	20		1	4
C	1	20		3
D	20	4	3	

Source: Modified from Bertsekas (2023)

*Example: Traveling Salesman Problem (Rollout at City B)



Rollout algorithm at AB
T_2



to

from	A	B	C	D
A		5	3	20
B	20		1	4
C	1	20		3
D	20	4	3	

Source: Modified from Bertsekas (2023)

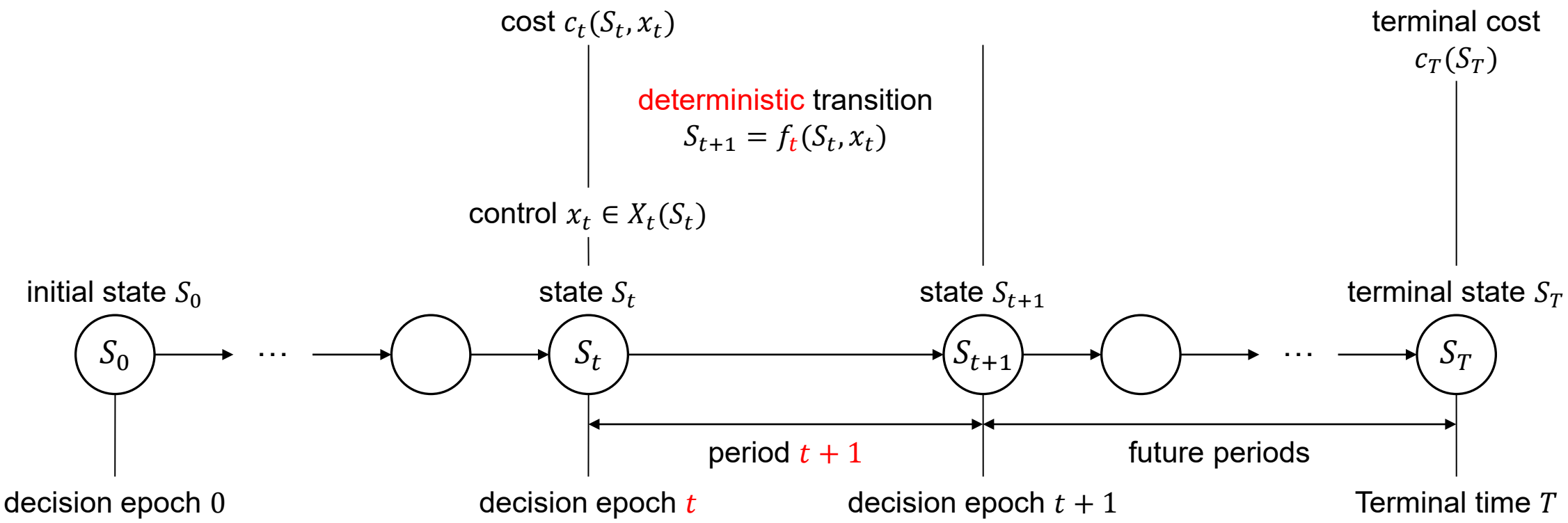
Deterministic Finite State Finite Horizon Problem

TOpS

- Notation and terminologies
  - t: time index (decision epoch or stage)
  - T: finite planning horizon, i.e., number of times control is applied
  - S_t: state of the system at time t
  - S_0: initial state of the system
  - x_t: control or decision variable of the system at time t, $x_t \in X_t(S_t)$, where $X_t(S_t)$ is the decision space
  - f_t: transition function (or system equation) that describes the mechanism by which the state is updated, $S_{t+1} = f_t(S_t, x_t)$
  - c_t: cost function, $c_t(S_t, x_t)$
  - c_N: terminal cost at the end of the horizon, $c_T(S_T)$
  - $V(S_0)$: value function (total cost) with respect to initial state S_0

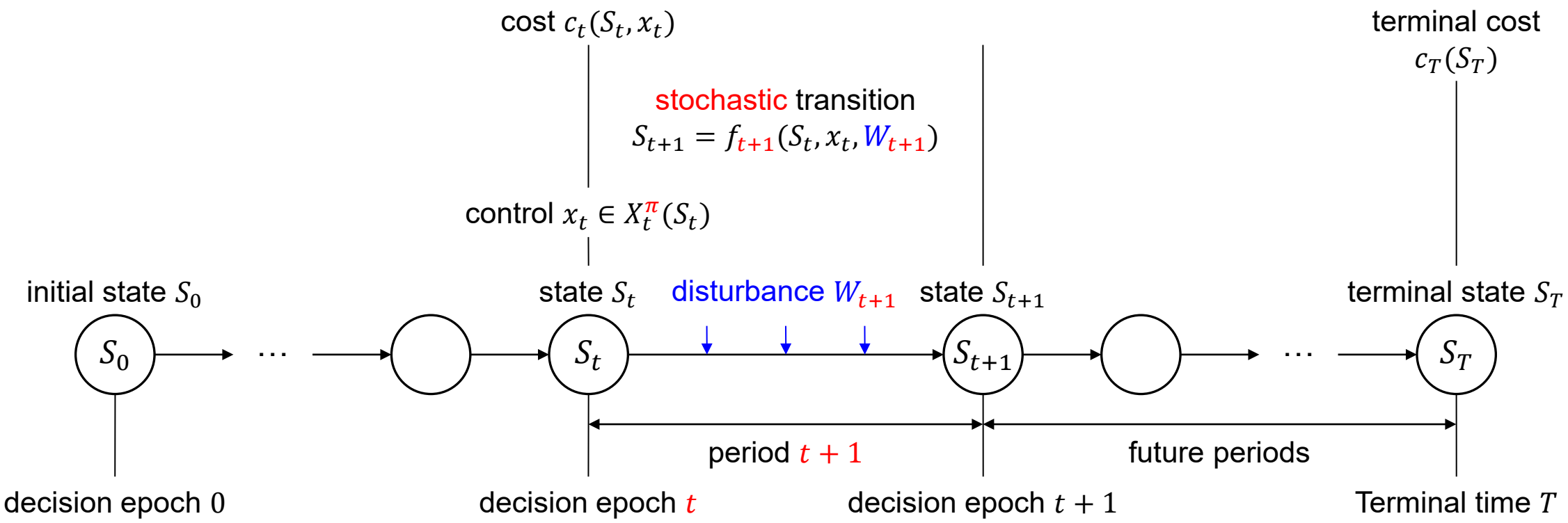
Stochastic Finite State Finite Horizon Problem

Deterministic Finite State Finite Horizon Problem (Recall)



Source: Modified from Bertsekas (2023) and Powell (2022)

*Stochastic Finite State Finite Horizon Problem



Source: Modified from Bertsekas (2023) and Powell (2022)

Deterministic Finite State Finite Horizon Problem (Recall)

- In finite horizon problems
  - A finite number of T time steps, indexed by time t (decision epochs or stages)
  - State of the system S_t at time t
  - Control or decision variable $x_t \in X_t(S_t)$ at time t, which depends on S_t
  - In a deterministic DP problem, the transition function (or system equation) is

$$S_{t+1} = f_t(S_t, x_t),\ t = 0, 1, \dots, T-1.$$

    - That is, the state S_{t+1} is determined solely by state S_t and decision x_t.
  - The system involves an additive cost function $c_t(S_t, x_t)$.
  - Given initial state S_0, the total cost of a control sequence $\{x_0, \dots, x_{T-1}\}$ is

$$V(S_0; x_0, \dots, x_{T-1}) = c_T(S_T) + \sum_{t=0}^{T-1} c_t(S_t, x_t),$$

    - where $c_T(S_T)$ is the terminal cost of terminal state S_T.
    - $V(S_0; x_0, \dots, x_{T-1})$ is well-defined, since $\{x_0, \dots, x_{T-1}\}$ and S_0 determines exactly $\{S_1, \dots, S_T\}$.
  - Minimize the total cost over all feasible sequences: optimal value as a function of S_0

$$V^*(S_0) = \min_{x_t \in X_t(S_t),\ t=0,\dots,T-1} V(S_0; x_0, \dots, x_{T-1}).$$

Source: Modified from Bertsekas (2023) and Powell (2022)

Stochastic Finite State Finite Horizon Problem

- In finite horizon problems
  - A finite number of T time steps, indexed by time t (decision epochs or stages)
  - State of the system S_t at time t
  - Control or decision variable $x_t \in X_t^\pi(S_t)$ at time t under policy π, which depends on S_t
  - In a stochastic DP problem, the transition function (or system equation) is

$$S_{t+1} = f_{t+1}(S_t, X_t^\pi(S_t), W_{t+1}),\ t = 0, 1, \dots, T-1.$$

    - That is, the state S_{t+1} is determined by state S_t, decision x_t, and random disturbance W_{t+1}.
  - The system involves an additive cost function $c_t(S_t, X_t^\pi(S_t), W_{t+1})$.
  - Given initial state S_0, the expected total cost under policy π is

$$V^\pi(S_0) = E_W[c_T(S_T) + \sum_{t=0}^{T-1} c_t(S_t, X_t^\pi(S_t), W_{t+1})],$$

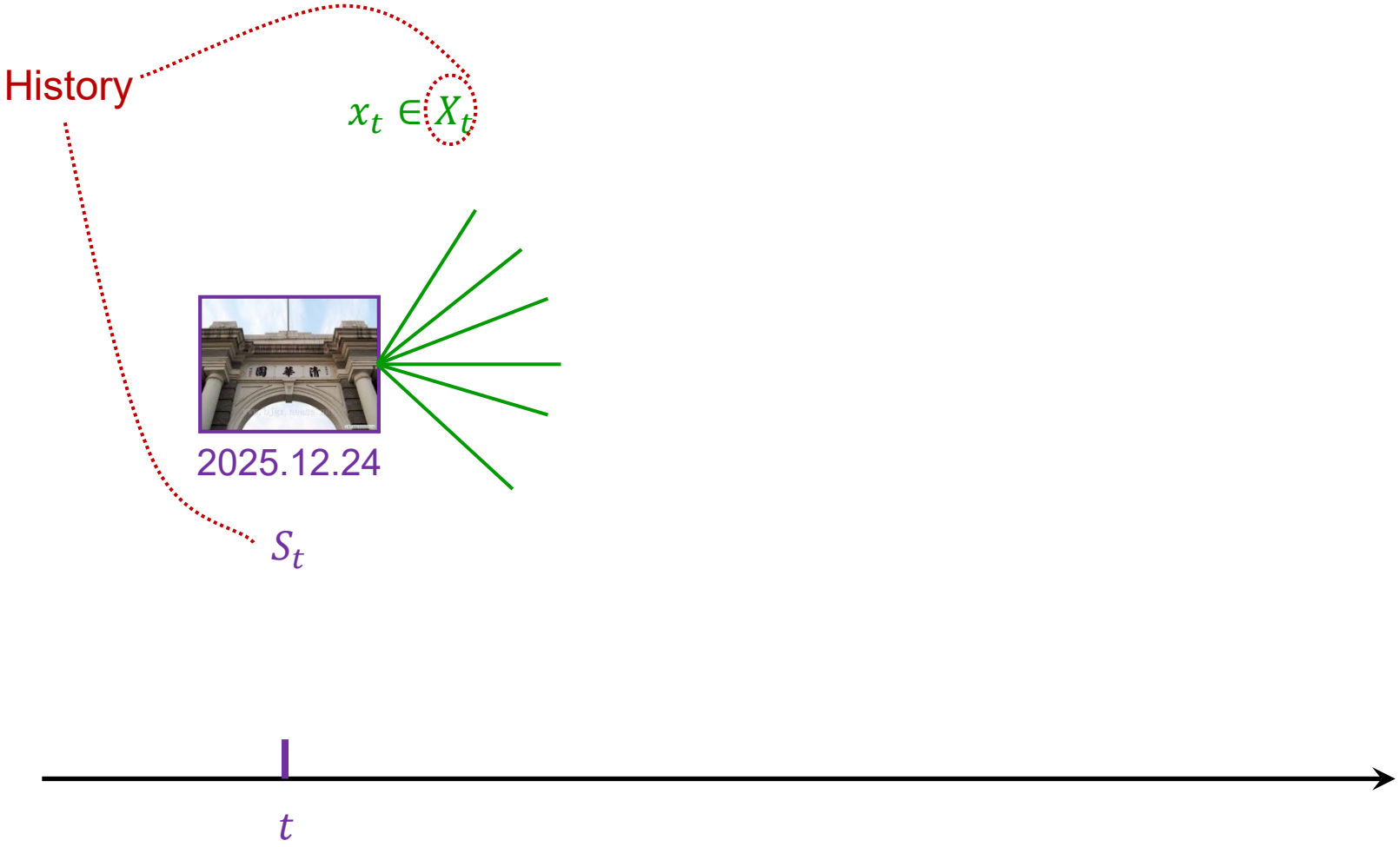
    - where $c_T(S_T)$ is the terminal cost of terminal state S_T.
  - Minimize the expected total cost over all policies: optimal value as a function of S_0

$$V^*(S_0) = \min_{\pi \in \Pi} V^\pi(S_0).$$

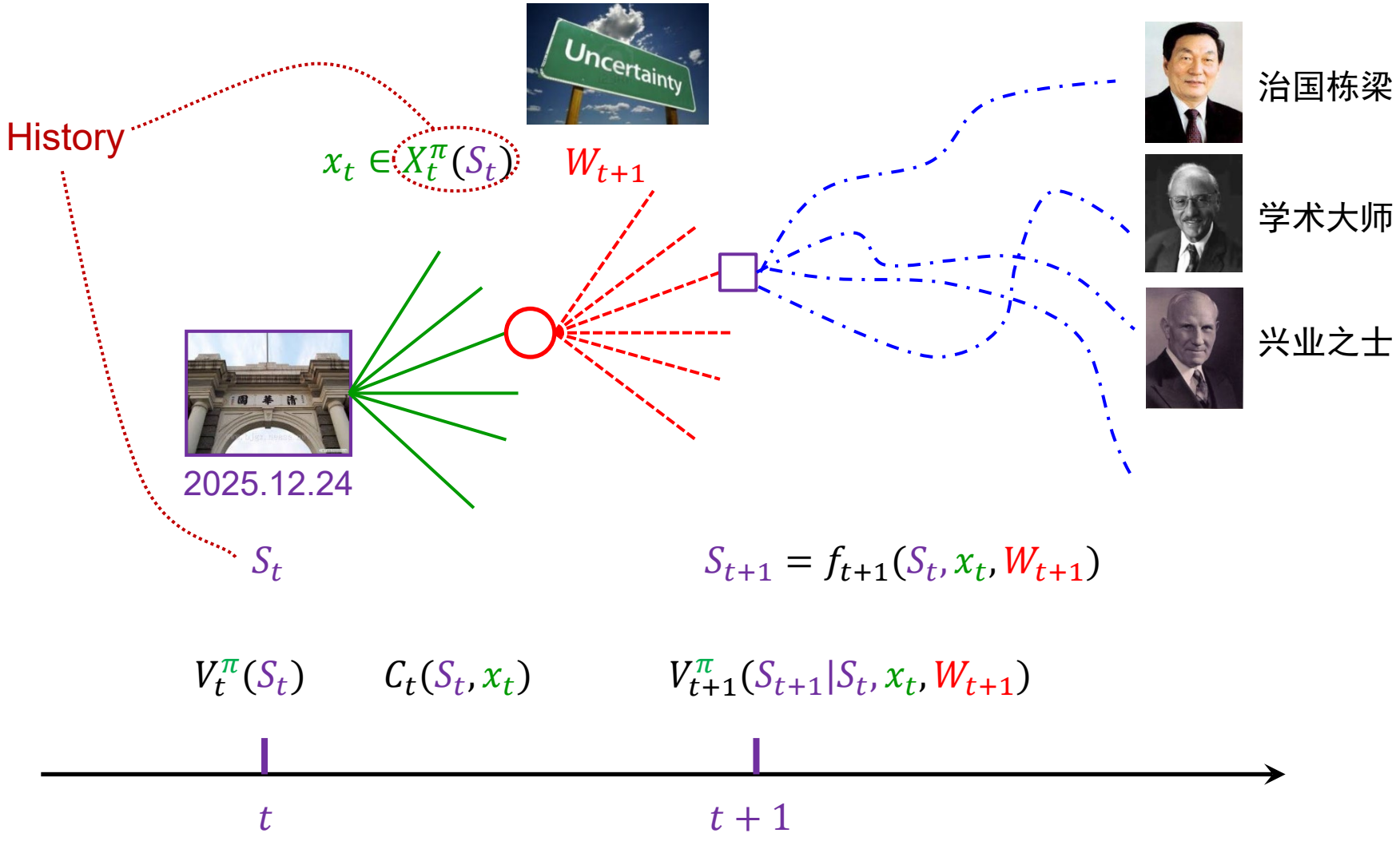
Source: Modified from Bertsekas (2023) and Powell (2022)

*Stochastic Optimization in Life

*Stochastic Optimization in Life



*Stochastic Optimization in Life



#Where to Go? 过去10年的主要就业方向

TOpS

- 行业
  - 信息传输、软件和信息技术服务业、教育、公共管理、社会保障和社会组织、科学研究和技术服务业、金融业、交通运输、仓储和邮政业、制造业……
- 单位
  - 中远海运、中外运、华能集团、中航工业、航天科技、中国商飞等国有企业
  - 清华、北大、人大、康奈尔、佐治亚理工、密歇根大学、南加大、华盛顿大学（西雅图）、约翰霍普金斯大学、帝国理工等国内外高校和科研院所
  - 国家教育部、发改委、科技部、北京市委组织部等中央及地方政府社会公共管理部门
  - 国家开发银行、进出口银行、招商银行、中信集团等金融类企业
  - 华为、腾讯、阿里、美团、京东等民营企业
- 境内外深造
  - 清华大学、北京大学、哈佛大学、麻省理工学院、斯坦福大学、加州大学伯克利分校、密歇根大学安娜堡分校、牛津大学、剑桥大学、帝国理工学院、慕尼黑工业大学、海德堡大学、苏黎世联邦理工学院、新加坡国立大学……

Where to Go? IE系友如是说

什么是工业工程？

IE 系友如是说——

学界篇

听系友们“原汁原味”的真实声音

彭小珊

2006年考入清华大学工业工程系，2017年获得芝加哥大学博士学位

就职单位及职务 | 单位：印第安纳大学凯利商学院 职务：助理教授

主要负责的工作 | 主要负责教学研究工作

我认为工业工程是工程学和管理学的最佳结合。本科四年学习既提供扎实的数理训练，又强调所学工具在各个行业的应用场景，为以后从事咨询，数据科学或者学术科研等多种职业方向都提供了良好的基础。

陈秋实

2006年考入清华大学工业工程系，2016年获得美国佐治亚理工大学博士学位

就职单位及职务 | 单位：宾夕法尼亚州立大学工业工程系 职务：助理教授

主要负责的工作 | 从事医疗决策建模和卫生政策分析的研究，指导博士生科研，研究生和本科生教学

我自己学习工业工程最大的收获是从系统出发的思维方式，以及对复杂系统的抽象建模的能力。工业工程的方法不限于制造业和工厂，也可以应用于许多其他需要科学决策分析和规划的系统（比如在物流，配送，服务，医疗等系统中的资源优化配置）。

王 和

2007年考入清华大学工业工程系，2016年获得麻省理工学院博士学位

就职单位及职务 | 单位：佐治亚理工学院 职务：助理教授

主要负责的工作 | 主要负责运筹学和供应链领域的科研、教学工作

我认为工业工程是结合数学、计算机、经济学和管理科学的方法解决企业实际中面临的大规模、高复杂度的决策问题。

什么是工业工程？

IE 系友如是说——

业界篇-第1期

听系友们“原汁原味”的真实声音

杨觅

2003年考入清华大学工业工程系，2009年获得哥伦比亚大学硕士学位

就职单位及职务 | 单位：滴滴出行 职务：拉美拓展总监

主要负责的工作 | 主要负责滴滴在拉丁美洲的业务拓展工作

我认为工业工程是通过定量的方法分析优化真实社会中的生产和服务业的系统工程。工业工程人才具有工程和管理科学的综合知识，理论加国内外实践的全面能力，是现代化社会中复杂系统工程与管理机构中的必备人才。

惠雷

2003年考入清华大学工业工程系，2009年获得清华大学硕士学位

就职单位及职务 | 单位：顺丰速运 职务：营运规划处 负责人

主要负责的工作 | 主要负责顺丰全网规划设计工作

我认为工业工程首先是一种思维方式，是用技术手段优化和解答现实问题的直觉反应，是深入扎根到具体现实问题的过程，是取舍和平衡的技巧，是获取和分配资源的艺术。

陈正超

2004年考入清华大学工业工程系，2008年获得清华大学学士学位

就职单位及职务 | 苹果采购运营管理（上海）有限公司 物料项目经理

主要负责的工作 | iPhone原材料的物料计划和库存控制

工业工程是一门理论联系实际的学科，在企业的日常运营管理中有着广泛的应用。掌握了工业工程理论知识的专业人员，可以帮助企业提高运营效率，显著降低运营成本。

什么是工业工程？

IE 系友如是说——

公共服务部门篇

听系友们“原汁原味”的真实声音

徐旭雁

2003年考入清华大学工业工程系，2009年获得清华大学硕士学位

就职单位及职务 | 单位：佛山市教育局 职务：副局长

主要负责的工作 | 主要负责各类教育管理工作

工业工程是一门关于包含人的系统的质量、效用和效率的工程学科。人类社会的进步需要管理创新，IE则是管理中的“硬核”部分。在公共服务的庞大系统中，非常需要有IE思维和IE能力的人才。提升系统绩效，我们会想到通过系统改善而不仅是增加人财物的投入。当然在公共服务中，需要更多考虑人文关怀和各种系统绩效之外的考量指标。

吴筱君

2007年考入清华大学工业工程系，2016年获得清华大学博士学位

就职单位及职务 | 单位：清华大学党委研究生工作部 职务：讲师

主要负责的工作 | 主要负责研究生思想政治工作与教育管理

我认为工业工程是一门管理学科，“优化”和“以人为本”是其最为突出的两个特点。无论是时间管理、人员管理、流程管理还是资源管理等，都可以运用工业工程的理念和方法，改善现有的工作，用更低的成本产生更好的效益；基于人的身理心理学特点来做管理不仅更人性化，而且能取得事半功倍的效果。

杨林栋

2009年考入清华大学工业工程系，2018年获得清华大学博士学位

就职单位及职务 | 单位：龙岩国家经济技术开发区 职务：管委会副主任

主要负责的工作 | 主要负责产业研究、招商引资、军民融合工作

IE是极具竞争力的综合性、管理型工程学科。四年的本科训练会让我们具备较好的数理和信息技术基础，不管是在制造业，还是咨询、金融等行业，都有独特的优势。我们不是行业导向型专业，而是技能导向型专业，工业工程大有可为，IE人前景广阔！

Source: IE系友如是说（2019/04/03）、IE系友如是说•第二弹（2019/04/09）

Q & A